

SUGGESTED READING**General**

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Technical

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PROBLEMS

- 27.1** By equating the period–luminosity relation, Eq. (14.1), and the period–luminosity–color relation, Eq. (27.1), estimate the range of the $B - V$ color index for Cepheid variable stars.
- 27.2** The central light ring produced by SN 1987A, shown in Fig. 15.14, gave astronomers a unique chance to determine its distance (and therefore the distance to the Large Magellanic Cloud). The supernova heated the ring of gas, causing it to glow. The ring, which has an angular diameter of $1.66''$ (long axis), is presumed to be circular but tilted from the perspective of Earth. Light was received from the near side of the ring 340 days before light arrived from its far side.
- Carefully measure the photograph, and determine the angle between the plane of the ring and the plane of the sky.
 - Use the time delay to determine the diameter of the ring in parsecs, and compare your result with the caption to Fig. 15.14.
 - Use trigonometry to find the distance to SN 1987A.
- 27.3** The three brightest red stars in the galaxy M101 (the “Pinwheel” galaxy; see Fig. 25.5) have visual magnitudes of $V = 20.9$. Assuming that there is 0.3 mag of extinction, what is the distance to M101? How does this compare to the distance of 7.5 Mpc found using classical Cepheids?
- 27.4** Show that an uncertainty in the distance modulus $m - M$ of 0.4 mag corresponds to an uncertainty in the distance of about 20%. What uncertainty in the distance modulus would produce a 5% uncertainty in the distance? What uncertainty in the distance modulus would produce a 50% uncertainty in the distance?
- 27.5** In Eq. (27.4), the value of C for the Fornax cluster of galaxies is $C = -1.264$. What is the ratio of the distances to the Virgo and Fornax clusters? to the Coma and Fornax clusters?
- 27.6** Use the solid line in Hubble’s velocity–distance diagram, Fig. 27.7, to determine his value of H_0 . Why was his result so different from today’s value?

- 27.7** Use the relative motion of the Andromeda and Milky Way galaxies to estimate their total mass for the case $h = 0.5$. What is the corresponding mass-to-light ratio of these galaxies?
- 27.8** The Magellanic Stream orbits the Milky Way and extends from 50 kpc to 100 kpc from the Galactic center.
- (a) Consider a clump of gas in the stream in a circular orbit about the Galactic center. Take the radius of the orbit to be 75 kpc and the orbital speed to be 244 km s^{-1} . Treating the Galaxy and the mass clump as point masses, estimate the mass of the Milky Way. What is the corresponding mass-to-light ratio?
 - (b) Suppose a clump of gas at the tip of the stream starts with zero radial velocity at a distance of 100 kpc and reaches a radial velocity of -220 km s^{-1} after falling down to 50 kpc from the Galactic center. Assuming that the transverse (orbital) velocity of the clump of gas has not changed, use conservation of energy to estimate the mass of the Milky Way. Find the corresponding mass-to-light ratio. As before, treat the Galaxy and the clump of gas as point masses.
- 27.9** Assuming that the Sculptor group of galaxies occupies a spherical volume of space, find the difference in magnitude between two identical objects located at the very front and back of the group.
- 27.10** Derive Eq. (27.17).
- 27.11** Suppose that the densities of the dark matter and interstellar gas in M87 have the same r -dependence (i.e., they are proportional). Use Eq. (27.17) to show that the gas is isothermal. *Hint:* Assume $T \propto r^\alpha$ and show $\alpha = 0$. Use flat rotation curves and make use of Eqs. (24.49) and (24.50). [The controversial assumption that $T(r) = \text{constant}$ has been used in some studies of dark matter in elliptical galaxies.]
- 27.12** Like the Coma cluster, the Virgo cluster contains a large amount of hot (70 million K) intra-cluster gas that emits X-rays.
- (a) If the X-ray luminosity of the Virgo cluster's intracluster gas is about $1.5 \times 10^{36} \text{ W}$, use Eq. (27.20) to find the electron number density and the mass of the gas. Assume that the Virgo cluster is a sphere of radius 1.5 Mpc that is filled with completely ionized hydrogen.
 - (b) Use $L_V = 1.2 \times 10^{12} L_\odot$ for the visual luminosity of the Virgo cluster to estimate the cluster's luminous mass. How does this compare with your answer to part (a) for the mass of the intracluster gas?
 - (c) Assuming that the gas has no energy source and that it is simply losing energy via thermal bremsstrahlung, use Eq. (10.17) for the average kinetic energy per gas particle (protons and electrons) to estimate how long it will take for the gas to lose all of its energy. (Assume that the X-ray luminosity remains constant throughout your calculation.) How does your answer compare with the Hubble time, t_H ?
- 27.13** Estimate how long a galaxy in the Coma cluster would take to travel from one side of the cluster to the other. Assume that the galaxy moves with a constant speed equal to the cluster's radial velocity dispersion. How does this compare with the Hubble time, t_H ? What can you conclude about whether the galaxies in the Coma cluster are gravitationally bound?
- 27.14** For the galaxies in the Virgo cluster, the dispersion in the radial velocity is $\sigma_r = 666 \text{ km s}^{-1}$. Use the virial theorem to estimate the mass of the Virgo cluster.
- 27.15** Suppose that the galaxies in the Coma cluster are all moving *away* from the cluster's center but that the measured value of the radial velocity dispersion is unchanged. (For example,

directing each galaxy's present velocity radially outward would accomplish this.) In this case, the cluster would not be in equilibrium. What is the sign of the term $\langle d^2I/dt^2 \rangle$ in the virial theorem, Eq. (2.44)? Explain how neglecting this term would affect your estimate of the mass of the Coma cluster. How would your answer be different if all of the galaxies were moving *toward* the cluster's center?

- 27.16** For the galaxy NGC 5585, the quantity $2v_r/\sin i = 218 \text{ km s}^{-1}$, and its apparent H magnitude is $H = 9.55$ (already corrected for extinction). Use the Tully–Fisher method to determine the distance to this galaxy.
- 27.17** The brightest galaxy in the cluster A1060 has an apparent visual magnitude of $V = 10.99$. Estimate the distance to the cluster. Use the uncertainty in the average absolute magnitude of the brightest galaxy to determine how far off your answer could be.
- 27.18** The Galactic coordinates of the direction of the large-scale streaming motion of galaxies in the direction of the Great Attractor are $\ell = 309^\circ$, $b = 18^\circ$. Convert these to equatorial coordinates, and use a star chart to confirm that your answer is in the constellation of Centaurus.

COMPUTER PROBLEM

- 27.19** Given the results in Table 27.1 for the distance to the Virgo cluster of galaxies, the problem arises of how d_{ave} , the average distance, should be chosen.

(a) d_{ave} might be chosen to minimize

$$\Delta \equiv \sum_i \left(\frac{d_i - d_{\text{ave}}}{\delta_i} \right)^2,$$

where δ_i is the uncertainty in the distance d_i determined by the i th method in the table (e.g., for the estimate using novae, $d = 21.1 \text{ Mpc}$ and $\delta = 3.9 \text{ Mpc}$). (There are echoes of the least-squares fit to a straight line in this approach.) Make a graph of Δ vs. d_{ave} for values of d_{ave} between 15 and 22 Mpc, and determine (to the nearest 0.1 Mpc) the value of d_{ave} that minimizes Δ .

(b) The *weighted mean* value is

$$d_w = \frac{\sum_i (d_i/\delta_i^2)}{\sum_i (1/\delta_i^2)}.$$

Calculate the weighted mean for the values in Table 27.1, and compare your answer with what you found in part (a).

(c) Prove that your answers to parts (a) and (b) will always agree.

CHAPTER

28

Active Galaxies

- 28.1 *Observations of Active Galaxies*
- 28.2 *A Unified Model of Active Galactic Nuclei*
- 28.3 *Radio Lobes and Jets*
- 28.4 *Using Quasars to Probe the Universe*

28.1 ■ OBSERVATIONS OF ACTIVE GALAXIES

The story of modern astrophysics is one of a dynamically evolving universe. On every scale, from planets to stars to galaxies, the objects that are present in this era differ from what they were during previous epochs. As we study the ancient light that arrives from distant corners of the universe, we are able to examine how galaxies looked and behaved in their youth. These observations reveal a level of activity in the centers of young, remote galaxies that is rarely found in nearer galactic nuclei.

Seyfert Galaxies

The first hint of the violent heritage of today's galaxies was found by Edward A. Fath (1880–1959), who in 1908 was observing the spectra of “spiral nebulae.” Although most showed an absorption-line spectrum produced by the combined light of the galaxy's stars, NGC 1068 displayed six bright emission lines. In 1926 Edwin Hubble recorded the emission lines of this and two other galaxies. Seventeen years later Carl K. Seyfert (1911–1960) reported that a small percentage of galaxies have very bright nuclei that are the source of broad emission lines produced by atoms in a wide range of ionization states. These nuclei are nearly stellar in appearance.

Today these objects are known as **Seyfert galaxies**, with spectra that are categorized into one of two classes. **Seyfert 1** galaxies have very broad emission lines that include both allowed lines (H I, He I, He II) and narrower forbidden lines (such as [O III]).¹ Seyfert 1 galaxies generally have “narrow” allowed lines as well, although even the narrow lines are broad compared to the spectral lines exhibited by normal galaxies. The width of the lines is attributed to Doppler broadening, indicating that the allowed lines originate from sources with speeds typically between 1000 and 5000 km s^{−1}, while the forbidden lines correspond to speeds of around 500 km s^{−1}. **Seyfert 2** galaxies have only narrow lines (both permitted

¹Recall from Section 11.2 that forbidden lines involve low-probability transitions in atoms and are an indication of low gas densities.

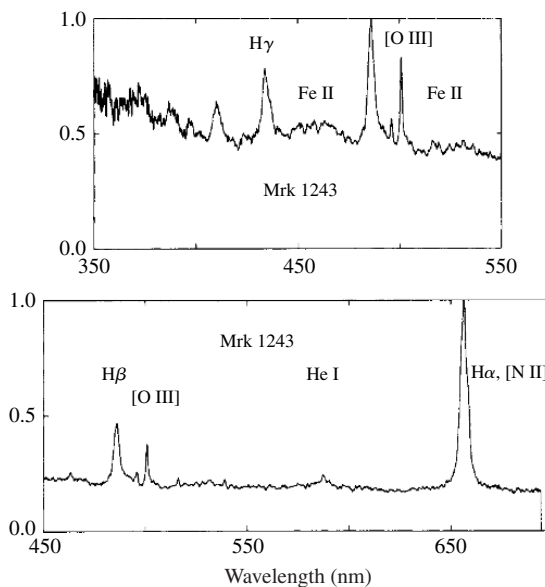


FIGURE 28.1 The visible spectrum of Mrk 1243, a Seyfert 1 galaxy. (Figure adapted from Osterbrock, *QJRAS*, 25, 1, 1984.)

and forbidden), with characteristic speeds of about 500 km s^{-1} . Every spectrum also shows a featureless continuum that is devoid of lines, originating from a small central source. The great luminosity of a Seyfert 1 galaxy arises from this continuum, which often overwhelms the combined light of all of the galaxy's stars. The continuum observed for a Seyfert 2 is significantly less luminous.

Figures 28.1 and 28.2, respectively, show the visible spectra of Mrk 1243 (a Seyfert 1) and Mrk 1157 (a Seyfert 2), where “Mrk” indicates an entry in the galaxy catalog of E. B. Markarian (1913–1985), produced in 1968. Some spectra display both broad and narrow permitted lines, and so they are classified as an intermediate type such as Seyfert 1.5. However, it is important to emphasize that this is a spectral classification. The spectra of a few Seyfert galaxies have changed nearly from type 1.5 to type 2 in a matter of years, although the broad $H\alpha$ emission line has rarely if ever completely disappeared.

The galaxies known to emit the most X-ray energy are Seyferts of types 1 and 1.5. The X-ray emission is quite variable, and can change appreciably on timescales ranging from days to hours. In contrast, X-rays are less frequently measured for Seyfert 2 galaxies. An analysis of the hard X-rays that are observed for Seyfert 2s indicates that the “missing” X-rays have been absorbed by intervening material with huge hydrogen column densities² of between 10^{26} and 10^{28} m^{-2} .

Seyferts make up only a few tenths of a percent of all field galaxies. It is interesting that at least 90% of the Seyferts close enough to be resolved by telescopes are spiral galaxies, typically of types Sb or SBb. They are frequently accompanied by other galaxies with which

²Recall from Section 9.5 that line profiles can be used to calculate the column density of the absorbing material.

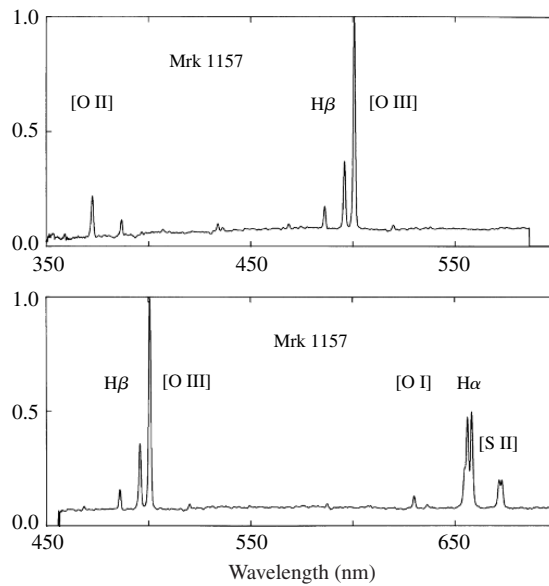


FIGURE 28.2 The visible spectrum of Mrk 1157, a Seyfert 2 galaxy. (Figure adapted from Osterbrock, *QJRAS*, 25, 1, 1984.)

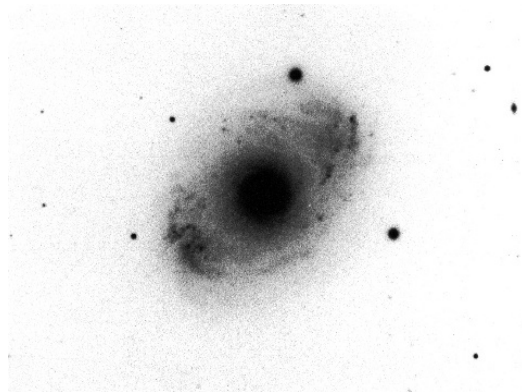


FIGURE 28.3 A long exposure of the Seyfert 1 (or 1.5) galaxy NGC 4151 showing the galactic disk around its bright nucleus. (Image from Sandage and Bedke, *The Carnegie Atlas of Galaxies*, Carnegie Institution of Washington, Washington, D.C., 1994.)

they may be gravitationally interacting. Figure 28.3 is a long-exposure view of the Seyfert galaxy NGC 4151 (type Sab) that shows the galactic disk around its bright nucleus.

The Spectra of Active Galactic Nuclei

Seyferts belong to the general class of galaxies with **active galactic nuclei**, or AGN for short. Other members of this class, such as radio galaxies, quasars, and blazars, will be introduced in the discussion that follows.

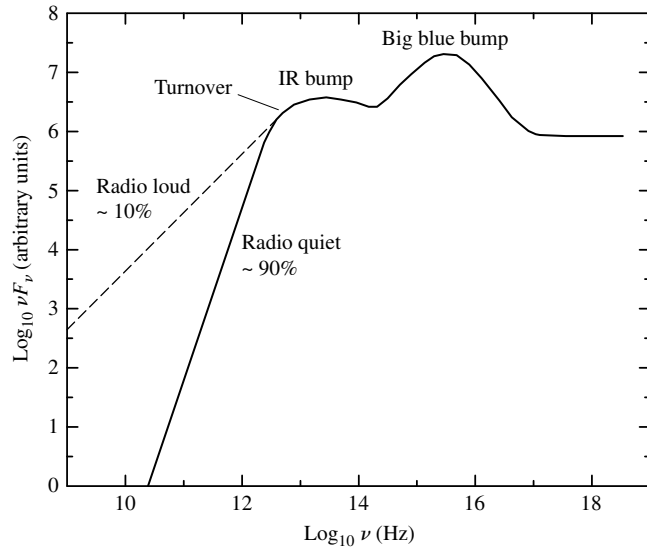


FIGURE 28.4 A sketch of the continuum observed for many types of AGNs.

Figure 28.4 is a rough schematic of the continuum observed for many types of AGNs (note that the logarithm of the product νF_ν is plotted on the figure's vertical axis). The most notable feature of this **spectral energy distribution** (SED) is its persistence over some 10 orders of magnitude in frequency. This wide spectrum is markedly different from the thermal (blackbody) spectrum of a star or the combined spectra of a galaxy of stars.

When AGNs were first studied, it was thought that their spectra were quite flat. Accordingly, a power law of the form

$$F_\nu \propto \nu^{-\alpha} \quad (28.1)$$

was used to describe the monochromatic energy flux, F_ν .³ The **spectral index**,⁴ α , was believed to have a value of $\alpha \simeq 1$.

The power received within any frequency interval between ν_1 and ν_2 is

$$L_{\text{interval}} \propto \int_{\nu_1}^{\nu_2} F_\nu d\nu = \int_{\nu_1}^{\nu_2} \nu F_\nu \frac{d\nu}{\nu} = \ln 10 \int_{\nu_1}^{\nu_2} \nu F_\nu d \log_{10} \nu, \quad (28.2)$$

so that equal areas under a graph of νF_ν vs. $\log_{10} \nu$ correspond to equal amounts of energy; hence the reason for plotting $\log_{10} \nu F_\nu$ on the ordinate in Fig. 28.4. A value of $\alpha \simeq 1$ reflects the horizontal trend seen to the right of the turnover in Fig. 28.4.

The continuous spectra of AGNs are now known to be more complicated, involving a mix of thermal and nonthermal emission. However, Eq. (28.1) is still used to *parameterize*

³Recall from Section 3.5 that $F_\nu d\nu$ is the amount of energy with a frequency between ν and $\nu + d\nu$ that arrives per unit area per second on a detector aimed at the source.

⁴Warning: Some authors define the spectral index with the opposite sign.

the continuum. The spectral index typically has a value between 0.5 and 2 that usually increases with increasing frequency, so the curve of $\log_{10} \nu F_\nu$ vs. $\log_{10} \nu$ in Fig. 28.4 is generally concave downward. In fact, the value of α is constant over only a limited range of frequencies, such as in the infrared and visible regions of the spectrum. The shape and polarization of the visible-UV spectrum indicates that it can sometimes be decomposed into contributions from thermal sources (blackbody spectrum, low polarization) and nonthermal sources (power-law spectrum, significant polarization). The thermal component appears as the **big blue bump** in Fig. 28.4, which can contain an appreciable amount of the bolometric luminosity of the source. It is generally believed that the emission from the big blue bump is due to an optically thick accretion disk, although some researchers have suggested that free-free emission may be responsible. Also evident is a thermal **infrared bump** to the left of the big blue bump; it is probably due to emission from warm ($T \lesssim 2000$ K) dust grains.

A pure power-law spectrum (with constant α) is the signature of synchrotron radiation, which is frequently encountered in astronomical situations involving relativistic electrons and magnetic fields (cf., Examples 4.3.3 and 16.7.1). As shown in Fig. 28.5, a synchrotron spectrum is produced by the combined radiation emitted by individual electrons as they spiral around magnetic field lines. If the distribution of the individual electron energies obeys a power law, then the resulting synchrotron spectrum is described by Eq. (28.1). However, the synchrotron spectrum does not continue to rise without limit as the frequency decreases. At a transition frequency, the spectrum turns over and varies as $\nu^{5/2}$ (spectral

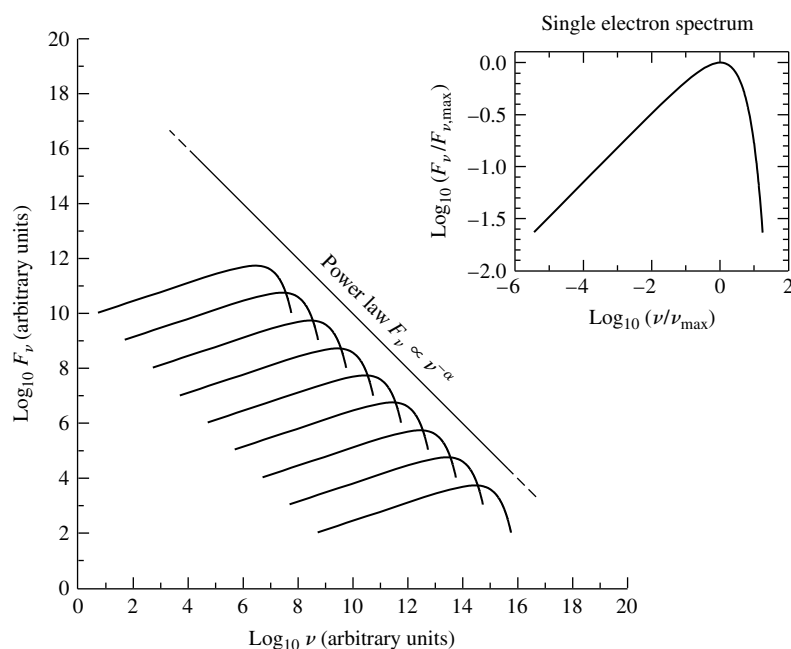


FIGURE 28.5 The power-law spectrum of synchrotron radiation, shown as the sum of the radiation produced by individual electrons as they spiral around magnetic field lines. The spectrum of a single electron is at the upper right. The turnover at low frequencies is not shown.

index $\alpha = -2.5$). This occurs because the plasma of spiraling electrons becomes opaque to its own synchrotron radiation, an effect known as **synchrotron self-absorption**.

In some SEDs, the “turnover” evident in the schematic continuum spectrum in Fig. 28.4 may be due to synchrotron self-absorption. However, the thermal contributions to the continuum spectrum evident in the infrared bump suggest that in other cases, the turnover may be due to the long-wavelength Rayleigh–Jeans portion of the blackbody spectrum produced by the warm dust grains. It is possible that the steeper, low-frequency spectra of radio-quiet AGNs are due to the thermal spectrum of dust grains, while the shallower, low-frequency spectra of radio-loud AGNs may be due to a combination of thermal and nonthermal emission.

Radio Galaxies

After World War II the science of radio astronomy that was started by Karl Jansky made rapid progress, led by astronomers in Australia and England. The first discrete source of strong radio waves (other than the Sun) was discovered in the constellation Cygnus and was named Cygnus A (a modern VLA radio image of Cyg A is shown in Fig. 28.6). Using the accurate position provided by English radio astronomer F. Graham Smith, the team of Walter Baade and Rudolph Minkowski (1895–1976) was able to find the optical counterpart of Cyg A. It is a peculiar-looking cD galaxy whose center is apparently encircled by a ring of dust (Fig. 28.7 shows an optical image of Cyg A obtained using the Hubble Space Telescope). Cyg A’s spectrum shows a redshift of $z = \Delta\lambda/\lambda_{\text{rest}} = 0.057$, corresponding to a recessional velocity of $16,600 \text{ km s}^{-1}$ (Eq. 4.38). From Hubble’s law in the form of Eq. (27.7), the distance to Cyg A is about $170h^{-1} \text{ Mpc}$ (implying a distance of 240 Mpc if $h = [h]_{\text{WMAP}}$). Considering that Cyg A is the brightest radio source beyond the Milky Way, this distance is surprisingly large. In fact, the only discrete radio sources brighter than

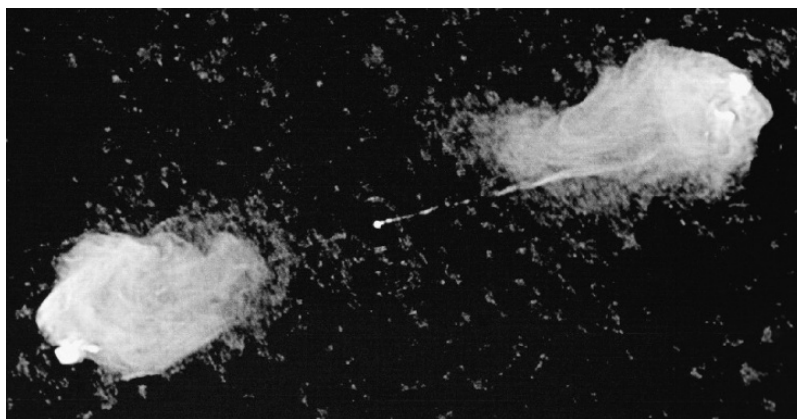


FIGURE 28.6 A VLA radio image of Cygnus A, showing the two radio lobes separated by about $100h^{-1} \text{ kpc}$ and the jet extending from the galaxy to the right-hand lobe. Cyg A is a narrow-line radio galaxy. The central cD galaxy does not show up on this radio picture (see Fig. 28.7). The width of the image is about 2 arcminutes. [Courtesy of R. A. Perley, J. W. Dreher, and J. J. Cowan (NRAO/AUI).]

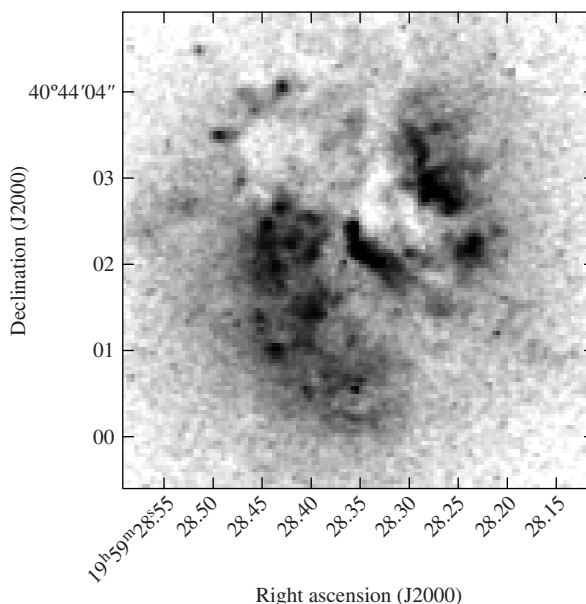


FIGURE 28.7 A continuum HST image of Cygnus A (3C 405) taken at a wavelength of 622 nm. (Figure adapted from Jackson, et al., *MNRAS*, 301, 131, 1998.)

Cyg A are the Sun and Cassiopeia A, the nearby (3 kpc) remnant of a Type II supernova. To be detected so far away, Cyg A must pour out enormous amounts of radio energy.

Cyg A is one example of a class of galaxies, called **radio galaxies**, that are extremely bright at radio wavelengths.

Example 28.1.1. The radio energy emitted by Cygnus A can be estimated by using its distance of $d = 170h^{-1}$ Mpc together with the observed value of the monochromatic flux at a radio frequency of 1400 MHz,

$$F_{1400} = 1.255 \times 10^{-23} \text{ W m}^{-2} \text{ Hz}^{-1} = 1255 \text{ Jy}.$$

The radio spectrum follows the power law of Eq. (28.1) with $\alpha \simeq 0.8$, so $F_\nu \propto \nu^{-0.8}$. That is, we can write

$$F_\nu = F_{1400} \left(\frac{\nu}{1400 \text{ MHz}} \right)^{-0.8}.$$

The radio luminosity can be found by integrating the monochromatic flux given by Eq. (3.29) over the range of radio frequencies found in Table 3.1. The upper frequency limit is taken to be $\nu_2 = 3 \times 10^9$ Hz, corresponding to a radio wavelength of 0.1 m. As is shown in Problem 28.20, the power-law behavior of the radio spectrum does not continue to $\nu = 0$. Instead, the flux received from Cygnus A declines when the frequency falls below about

continued

$\nu_1 = 10^7$ Hz. With these limits, the radio luminosity is approximately

$$L_{\text{radio}} = 4\pi d^2 \int_{\nu_1}^{\nu_2} F_\nu d\nu = 4\pi d^2 F_{1400} \int_{\nu_1}^{\nu_2} \left(\frac{\nu}{1400 \text{ MHz}} \right)^{-0.8} d\nu = 2.4 \times 10^{37} h^{-2} \text{ W}.$$

Using the WMAP value of $[h]_{\text{WMAP}} = 0.71$, the radio luminosity of Cygnus A is estimated to be $L_{\text{radio}} = 4.8 \times 10^{37} \text{ W}$. This is several million times more radio energy than is produced by a normal galaxy such as M31 and is roughly three times the energy produced at *all* wavelengths by the Milky Way.

Like Seyfert galaxies, radio galaxies may also be divided into two classes: **broad-line radio galaxies** (BLRGs, corresponding to Seyfert 1s) and **narrow-line radio galaxies** (NLRGs, corresponding to Seyfert 2s). BLRGs have bright, starlike nuclei surrounded by very faint, hazy envelopes. NLRGs, on the other hand, are giant or supergiant elliptical galaxies (types cD, D, and E); Cyg A is a NLRG.

Despite their similarities, there are obvious differences between Seyferts and radio galaxies. Although Seyfert nuclei emit some radio energy, they are relatively quiet at radio wavelengths compared with radio galaxies. Furthermore, while nearly all Seyferts are spiral galaxies, none of the strong radio galaxies are spirals.

Radio Lobes and Jets

A radio galaxy may display extended **radio lobes**, as in Fig. 28.6, or it may radiate its energy both from a compact **core** in its nucleus and from a **halo** that is about the size of the visible galaxy or larger. The optical cD galaxy in Fig. 28.7 is the central dot in the Cyg A radio image shown in Fig. 28.6. The optical galaxy is flanked by two huge radio lobes that are the sources of the tremendous radio luminosity estimated in Example 28.1.1. Each of the lobes has a diameter of about $17h^{-1}$ kpc.

Observations reveal that one of the lobes is connected to the central galaxy of Cyg A by a collimated **jet** that spans the roughly $50h^{-1}$ kpc of space separating the galaxy from the lobe. (Since the orientations of jets and radio lobes are not well determined, the values for their sizes quoted here are projected distances on the plane of the sky.) At least half of the stronger radio galaxies have detectable jets, as do more than three-quarters of the weaker sources. The jets associated with the powerful sources tend to be one-sided (like Cyg A's), while those found in less luminous radio galaxies are typically two-sided. One reason for this is that the stronger radio galaxies can be seen at greater distances, and so a dim counterjet may go undetected. (Reasons why counterjets may not appear as luminous to the observer will be discussed in Section 28.3.)

Figure 28.8 shows the strong jet and weak counterjet of the elliptical galaxy NGC 6251 at several radio frequencies. Note that the Moon's angular diameter would just fit into either of the two rectangular boxes at the top of the figure, while the scale of the bottom box is just a few milliarcseconds across. It is remarkable that the jet can be traced essentially all the way to the core of the galaxy.

Other radio jets are not as straight as those of Cyg A or NGC 6251. Figure 28.9 shows the windblown appearance of the jets emanating from NGC 1265, produced by that galaxy's motion through the intracluster gas of the Perseus cluster.

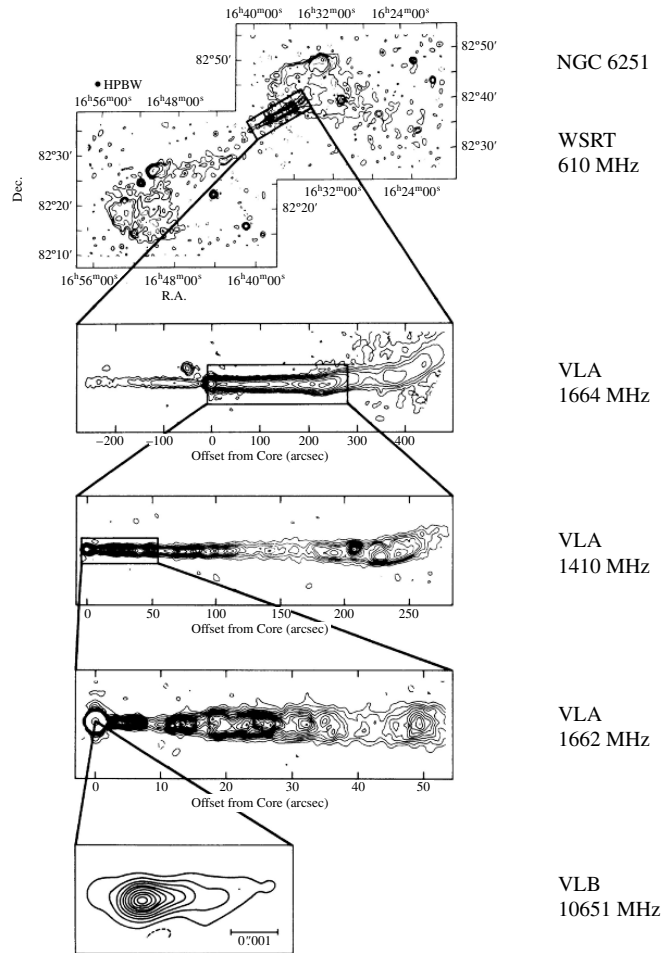


FIGURE 28.8 The jet and counterjet (second panel) of the radio galaxy NGC 6251. (Figure adapted from Bridle and Perley, *Annu. Rev. Astron. Astrophys.*, 22, 319, 1984. Reproduced by permission from the Annual Review of Astronomy and Astrophysics, Volume 22, ©1984 by Annual Reviews Inc.)

Following Cyg A, many more radio galaxies were discovered. One of these is M87, the giant elliptical (E1) galaxy described in Section 27.3 that lies at the center of the Virgo cluster. With an apparent visual magnitude of $V = 8.7$, M87 is one of the brighter-appearing galaxies in the sky. Figure 28.10 shows two HST views of M87, also known as Virgo A to radio astronomers. Its prominent jet, shown at the right, was discovered optically in 1917. The jet extends from the galaxy some 1.5 kpc into one of its radio lobes. The jet also displays evenly spaced knots that are bright at radio, visible, and X-ray wavelengths. The X-ray luminosity of M87, including the jet, is roughly 10^{36} W. This is about 50 times greater than M87's radio luminosity. The inset shows the spiral-shaped disk of hot gas that is at the core of M87. There is also evidence for a faint counterjet extending away from M87 in the direction opposite that of the dominant jet.

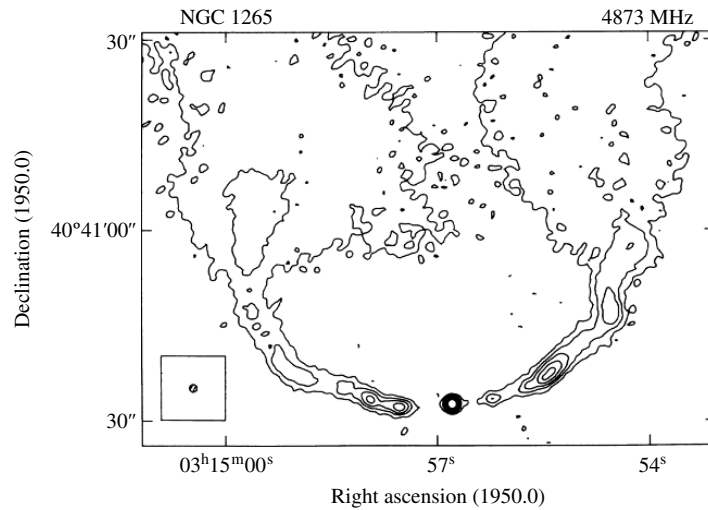


FIGURE 28.9 The radio appearance of NGC 1265, with its jets swept back by that galaxy's motion through the surrounding intracluster gas. (Figure adapted from O'Dea and Owen, *Ap. J.*, 301, 841, 1986.)

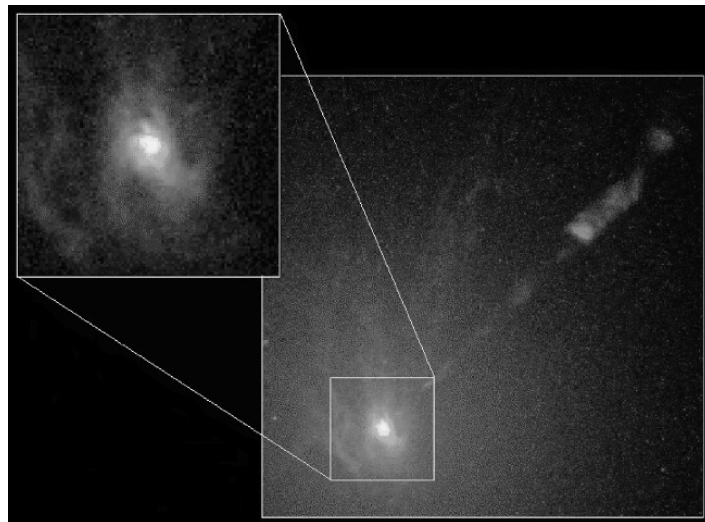


FIGURE 28.10 Two HST views of M87 and its jet. The inset shows the spiral-shaped disk of hot gas at the center of M87. [Courtesy of H. Ford (STScI/Johns Hopkins U.); R. Harms (Applied Research Corp.); Z. Tsvetanov, A. Davidsen, and G. Kriss (Johns Hopkins U.); R. Bohlin and G. Hartig (STScI); L. Dressel and A. K. Kochhar (Applied Research Corp.); and Bruce Margon (U. Washington).]

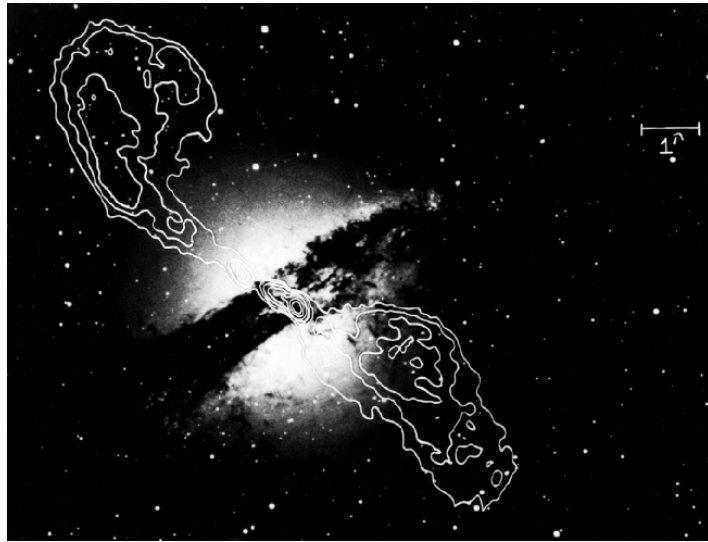


FIGURE 28.11 The visual and radio appearance (superimposed contour lines) of the radio galaxy Centaurus A. (Courtesy of NRAO.)

One of the largest radio galaxies known is 3C 236 (the “3C” designates a listing in the **Third Cambridge Catalog** of radio sources). With a redshift of $z = 0.0988$, its distance is about $280h^{-1}$ Mpc, according to Hubble’s law. The radio lobes of 3C 236 are separated by more than $1.5h^{-1}$ Mpc, projected onto the plane of the sky, while its radio jet is only $400h^{-1}$ pc long.

The closest example of an AGN is Centaurus A (NGC 5128), at a distance of $4.7h^{-1}$ Mpc. Figure 28.11 shows an optical image of Cen A, an E2 galaxy girded by a thick dust lane. Superimposed on the photograph is a radio map showing the radio lobes. (This figure corresponds roughly to the central region of Fig. 6.19.) Like M87, Cen A has a jet extending from its nucleus containing several knots of radio and X-ray emission. Although Cen A is in our astronomical backyard, radio galaxies on average are roughly 100 times less abundant than Seyferts in regions that are nearby in cosmological terms.

The Discovery of Quasars

As radio telescopes discovered increasing numbers of radio sources in the late 1950s, the task of identifying these sources with known objects became more important. In 1960 Thomas Matthews and Allan Sandage were searching for an optical counterpart to another radio source, 3C 48. They found a 16th-magnitude starlike object whose unique spectrum displayed broad emission lines that could not be identified with any known element or molecule. In Sandage’s words, “The thing was exceedingly weird.” In 1963 a similarly weird spectrum was found for another radio source with a stellar appearance, 3C 273. Figure 28.12 shows 3C 273 and its jet, which extends a projected distance of $39h^{-1}$ kpc from the nucleus.

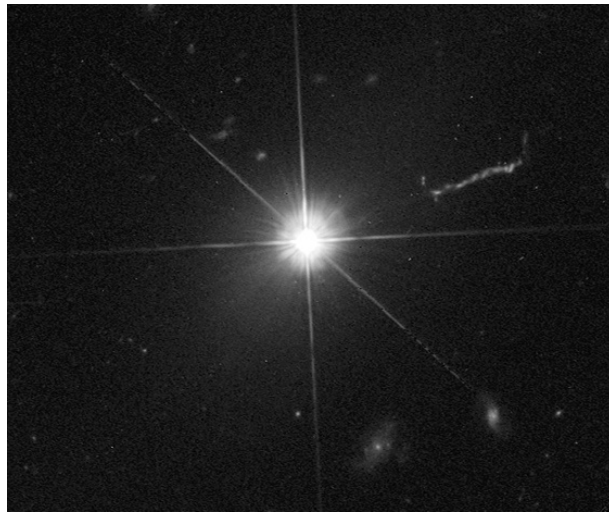


FIGURE 28.12 The quasar 3C 273 and its jet. [Courtesy of NASA and J. Bahcall (IAS).]

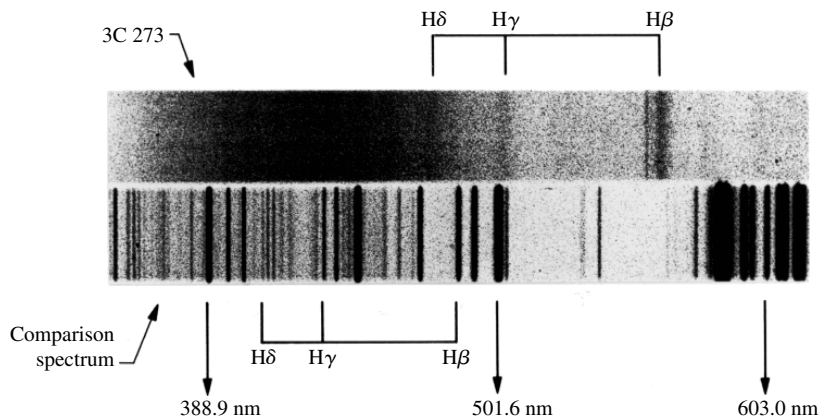


FIGURE 28.13 The $z = 0.158$ redshift of the quasar 3C 273. (Adapted from a figure courtesy of Palomar/Caltech.)

3C 48, 3C 273, and other, similar sources were classified as **quasi-stellar radio sources** (QSRs), which became known as **quasars**. But names are not a substitute for understanding, and an understanding of quasars proved to be elusive. Later that year, the mystery lifted somewhat when Dutch astronomer Maarten Schmidt recognized that the pattern of the broad emission lines of 3C 273 was the same as the pattern of the Balmer lines of hydrogen. These familiar lines had been severely redshifted ($z = 0.158$) to unfamiliar wavelengths, making their identification difficult; see Fig. 28.13. This Doppler shift means that 3C 273 is receding from Earth at 14.6% of the speed of light. According to Hubble's law, this places 3C 273 at a distance of about $440h^{-1}$ Mpc. At Caltech, Schmidt's associates Jesse

Greenstein (1909–2002) and Thomas Matthews calculated that 3C 48 has an even greater redshift, $z = 0.367$, corresponding to a radial velocity of $0.303c$ and a Hubble distance of just over $900h^{-1}$ Mpc. Astronomers realized that 3C 48 was one of the most distant objects yet discovered in the universe.

Quasar Luminosities

A quasar's radio emission may come either from radio lobes or from a central source in its core. Quasars are so far away that in optical images most appear as overwhelmingly bright, starlike nuclei surrounded by faint fuzzy halos. In some cases, a fuzzy halo can be resolved into a faint parent galaxy. To be visible from such great distances, quasars must be exceptionally powerful.

Example 28.1.2. Equation (3.6) can be used to obtain the absolute visual magnitude of the quasar 3C 273, which has an apparent visual magnitude of $V = 12.8$. Adopting $[h]_{\text{WMAP}} = 0.71$ yields a distance of $d \simeq 620$ Mpc, implying that

$$M_V = V - 5 \log_{10} \left(\frac{d}{10 \text{ pc}} \right) = -26.2.$$

This value can be used to obtain an estimate of the luminosity of the quasar at visual wavelengths. Using $M_{\text{Sun}} = 4.82$ for the Sun's absolute visual magnitude, Eq. (3.7) gives an estimate of the quasar's visual luminosity:

$$L_V \approx 100^{(M_{\text{Sun}} - M_V)/5} L_{\odot} = 2.6 \times 10^{12} L_{\odot} = 1 \times 10^{39} \text{ W}.$$

The radio energy emitted by 3C 273 can be estimated from its distance and the value of the monochromatic flux at a radio frequency of 1400 MHz, $F_{1400} = 4.64 \times 10^{-25} \text{ W m}^{-2} \text{ Hz}^{-1} = 46.4 \text{ Jy}$. The radio spectrum follows the power law of Eq. (28.1) with a spectral index of $\alpha \simeq 0.24$. Integrating the monochromatic flux from $\nu_1 \simeq 0$ to $\nu_2 = 3 \text{ GHz}$ gives

$$L_{\text{radio}} = 4\pi d^2 \int_{\nu_1}^{\nu_2} F_{\nu} d\nu = 7 \times 10^{36} \text{ W}.$$

The bolometric luminosities inferred for quasars range from about 10^{38} W to more than 10^{41} W , with $5 \times 10^{39} \text{ W}$ being a typical value; see Fig. 28.16. This implies that the most luminous quasars are on the order of 10^5 times more energetic than a normal galaxy like our own Milky Way.

Quasar Spectra

The monochromatic flux of 3C 273 is shown in Fig. 28.14. This continuous spectrum spans nearly 15 orders of magnitude in frequency, very broad compared with the sharply peaked blackbody spectrum of a star. The gentle decline at the low-frequency end of the spectrum reflects the larger-than-average spectral index ($\alpha = 0.24$) for 3C 273 in this regime. (At low frequencies, the spectrum of 3C 273 is dominated by radiation from its jet rather than

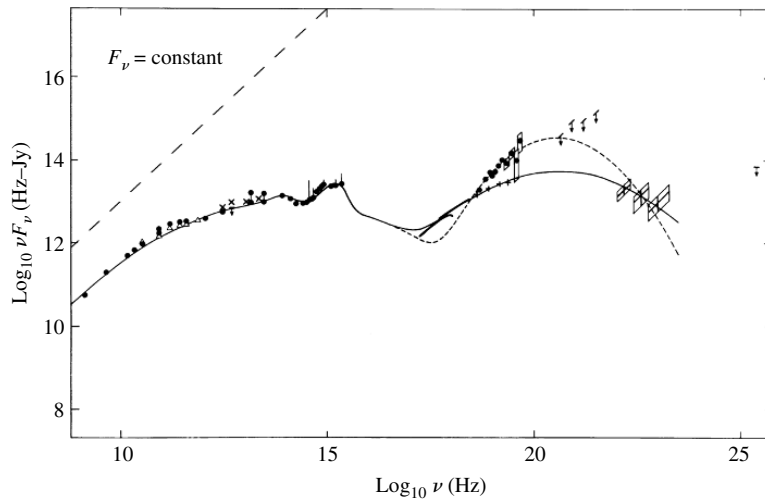


FIGURE 28.14 The spectrum of 3C 273, after the Doppler shift of the frequencies due to the Hubble flow has been removed. A horizontal line would correspond to a spectral index of $\alpha = 1$; for reference, the diagonal dashed line shows the slope for $F_\nu = \text{constant}$. The two lines on the right correspond to 3C 273 during quiescence and during an outburst. (Figure adapted from Perry, Ward, and Jones, *MNRAS*, 228, 623, 1987.)

from its core.) For most other quasars, the spectrum at the low-frequency end falls off more abruptly (smaller α). A typical spectrum turns over in the far infrared at a frequency of about 5×10^{12} Hz, possibly due to dust and/or synchrotron self-absorption. Also, although some quasars are most luminous at infrared wavelengths and others peak in X-rays, the peak power output of 3C 273 is in the form of low-energy gamma rays.

Quasars emit an excess of ultraviolet light relative to stars and so are quite blue in appearance. For example, the color indices of 3C 48 are $U - B = -0.61$ and $B - V = 0.38$ (you should note that this lies well above the positions of main-sequence stars on the color-color diagram in Fig. 3.11). In Fig. 28.14, this ultraviolet excess is indicated by the big blue bump between roughly 10^{14} Hz and 10^{16} Hz. A big blue bump is a feature of most (but not all) quasar spectra.

Absorption lines may also be present in some quasar spectra. In particular, Doppler-broadened absorption lines, found in up to 10% of the spectra of quasars, originate from sources with speeds exceeding 10^4 km s $^{-1}$. These lines are believed to be associated with the quasar itself. Many additional narrow absorption lines are typically seen in the spectra of quasars with high redshifts ($z > 2.2$) due to the Lyman series of hydrogen (see Section 5.3) and metals such as C IV and Mg II. These lines would normally appear at ultraviolet wavelengths but have been redshifted into the visible spectrum by the recessional velocity of the absorbing material. The absorption lines of a given quasar can be placed into different groups that share common redshifts. Furthermore, the redshifts of these narrow absorption lines are nearly always less than the redshift of the quasar's emission lines. The various groupings of lines are thought to arise from clouds of intervening material that lie between the quasar and Earth, as will be discussed in Section 28.4.

Quasi-Stellar Objects

The distinctive appearance of quasars, starlike with an excess of ultraviolet light, led astronomers to search for more objects fitting this description. In fact, choosing those objects with $U - B < -0.4$ results in a nearly complete list of possible quasars (those at very high z are redder), which must then be confirmed by a spectroscopic analysis. Researchers discovered that about 90% of the confirmed quasar candidates, and AGNs in general, are relatively radio-quiet. For this reason, most of these objects are technically referred to as **quasi-stellar objects** (QSOs), rather than quasars (QSRs).

Quasar Terminology

Today, the term *quasar* has come to be used almost universally for both radio-loud QSRs and radio-quiet QSOs. As a result, it is common to encounter the descriptions *radio-loud quasars* and *radio-quiet quasars*. However, it is also sometimes the case that QSO is used as an abbreviation for *quasar*. The terminology can be confusing in the literature, so it is important for you to understand the context in which the term is being used. In this text, we will generally use *quasar* to represent both classes of objects, making the distinction between radio-loud and radio-quiet when necessary.⁵

Ultraluminous Infrared Galaxies

Nearly all quasars have spectra similar to those of broad-line radio galaxies and Seyfert 1s, with bright power-law continua and broad emission lines (both allowed lines and narrower forbidden lines). Seyfert 2 spectra, with their narrow emission lines (both permitted and forbidden), appear to have no counterparts among the quasars. However, some astronomers argue that a subset of the galaxies that were cataloged by the IRAS satellite as being ultraluminous at infrared wavelengths, known as **ultraluminous infrared galaxies** (ULIRGs), should be considered quasars of type 2 rather than starburst galaxies; recall Section 26.1. It is suggested that the infrared light results from dust that absorbs and reradiates the light from the quasar nucleus.

The High Cosmological Redshifts of Quasars

The **Sloan Digital Sky Survey** (SDSS) has cataloged 46,420 quasars. The brightest entry in the catalog in the i band (centered on a wavelength of 748.1 nm) is the object SDSS 17100.62+641209.0 at a redshift of $z = 2.7356$, having $M_i = -30.242$. The most distant quasar in the catalog is SDSS 023137.65–072854.4 at a redshift of $z = 5.4135$, implying a recessional velocity of more than $0.95c$. In fact, there are 520 quasars in the SDSS catalog with redshifts greater than $z = 4$.

For such large cosmological redshifts, we must abandon using the Hubble law in the forms of Eqs. (27.7) and (27.8) to determine distances. Cosmological redshifts are caused by the expansion of the space through which the light travels, so for extremely large distances the total elongation of the wavelength depends on how the expansion of the universe has

⁵It has been pointed out by more than one astronomer that this confusing terminology is also a bit contradictory; to say that a particular quasar is radio-quiet is equivalent to saying that we are discussing a *radio-quiet* quasi-stellar *radio source* (based on the original definition of *quasar*)!

changed with time. The rate of expansion is changing in response to all of the matter and energy in the universe. For this reason, it is customary to quote the redshift, z , rather than an actual distance determination. You should keep in mind, however, that the fractional change in wavelength for a cosmological redshift is the same as the fractional change in the size of the universe, R , since the time when the light was emitted.⁶ That is,

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{emitted}}}{\lambda_{\text{emitted}}} = \frac{R_{\text{obs}} - R_{\text{emitted}}}{R_{\text{emitted}}},$$

which gives

$$\boxed{\frac{R_{\text{obs}}}{R_{\text{emitted}}} = 1 + z.} \quad (28.3)$$

Thus a redshift of $z = 3$ means that the universe is now four times larger than when the light was emitted.

Evidence for Quasar Evolution

As was pointed out in Section 26.2, the regions that are cosmologically close to us make up “today’s universe.” In observing these regions, astronomers can study galaxies as they appear here and now in the present epoch. When looking deeper into the universe, however, we see ancient photons that have been traveling for a long time from a source that may have changed significantly since the light began its journey. This essentially means that looking farther into space implies peering farther into the past. Thus telescopes serve as time machines, providing a window to the early universe.

Bright quasars were certainly more common at earlier epochs than they are now, as evidenced by observations at large and small z , respectively. Several factors could contribute to the greater space density of luminous quasars in the past. Both the total number of quasars and their luminosities may have been different then, and it is obviously a difficult task to disentangle these influences. A further complication is introduced by the expansion of the universe. The universe is larger today than it was at a redshift z by a factor of $1 + z$, so the space density of quasars would be greater in the past even if their numbers and luminosities have remained constant. To avoid unnecessary confusion caused by the expansion of the universe, astronomers have defined a **comoving space density** that mathematically removes the effect of the expanding universe. The number of objects per Mpc^3 at a redshift z is divided by $(1 + z)^3$, scaling the space density down to the value it would have today (at $z = 0$). The comoving space density of a constant number of nonevolving objects does not change as the universe expands, and so a change in this density implies that the number of objects is varying or that the objects are evolving (or both).

Statistical studies indicate that there are more than 1000 times as many quasars per Mpc^3 (comoving space density) brighter than $M_B = -25.9$ at $z = 2$ than there are today ($z = 0$). However, there is strong evidence that the total number of quasars has *not* changed significantly from the present ($z = 0$) back to roughly $z = 2$. Figure 28.15 shows several

⁶This fractional change can be measured by the average separation of its constituents, for instance. The meaning of “the size of the universe” will be made more precise in Section 29.1.

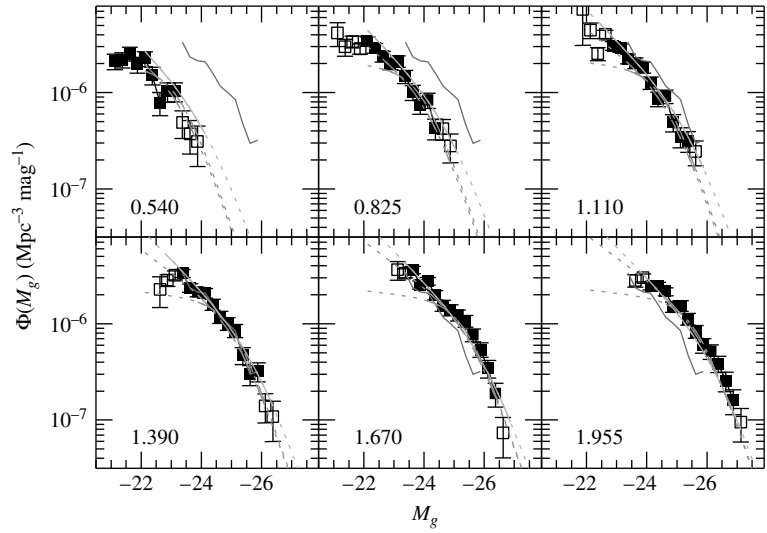


FIGURE 28.15 Luminosity functions for quasars with different redshifts. The redshifts are indicated in each frame; for instance, in the upper-left-hand frame, $z = 0.540$. The jagged line that is present in every frame (upper right in $z = 0.540$) represents the data for $z = 1.390$. Note that the population is brighter at greater redshifts. Data are from the Sloan Digital Sky Survey (SDSS) and the 2-degree Field survey (2dF) for 5645 quasars. The g band is centered on 480 nm. (Figure adapted from Richards et al., *MNRAS*, 360, 839, 2005.)

luminosity functions, Φ , for quasars in different redshift intervals, where $\Phi(M_g)$ is the number of quasars per Mpc^{-3} (comoving) that have an absolute magnitude between M_g and $M_g + dM_g$. Note that for $z < 2$, the curves would overlap if they were shifted horizontally along the M_g -axis. This indicates that for $z < 2$, the populations of quasars with different redshifts differ only in their luminosities, not in their comoving space densities. If this is so, then the scarcity of bright quasars today is an evolutionary effect, caused by a decrease in their luminosities with time. This luminosity evolution of quasars is shown in Fig. 28.16.⁷ Apparently, a picture in which a constant number of quasars grows dimmer as the universe expands is consistent with the observations for $z < 2$.

The situation becomes more complicated between $z = 2$ and $z = 3$. Astronomers can study the birth and evolution of quasars out to $z \sim 6$. Statistical surveys at both optical and X-ray wavelengths show that the comoving space density of AGNs peaks at a redshift of approximately $z \approx 2.5$ and then drops off for $z > 3$; see Fig. 28.17. These studies indicate that the comoving space density declines by roughly a factor of 10 from its peak value by $z \approx 4$.

This high- z deficit in the number of quasars could reflect a growth phase of supermassive black holes that power the nascent AGN. You may recall from Fig. 25.15 that a well-defined relationship exists between the mass of a supermassive black hole and the velocity dispersion of the spheroid of a galaxy, suggesting that as the mass of the galaxy grows and the velocity

⁷Note that Figure 28.16 assumes a pre-WMAP value of $h = 0.5$ and a specific model for the expansion of the universe that corresponds to a “flat universe” (a deceleration parameter of $q_0 = 0.5$; see Eq. 29.56).

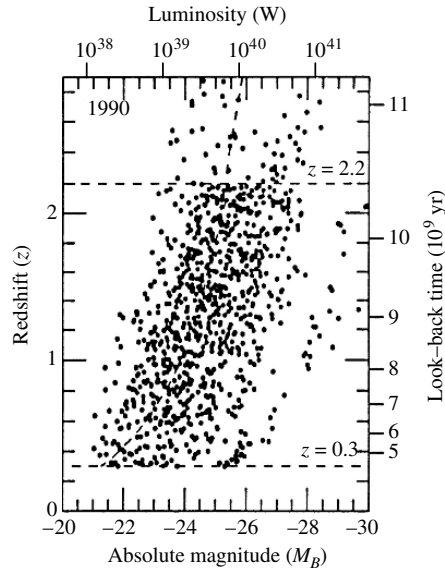


FIGURE 28.16 The dimming of quasars with time. For $z < 0.3$, there are too few nearby objects to provide an adequate sample for this figure. The empty region at the upper left has not been sampled by observations in this study. [Figure adapted from Boyle, *The Environment and Evolution of Galaxies*, Shull and Thronson (eds.), Kluwer Academic Publishers, Dordrecht, 1993.]

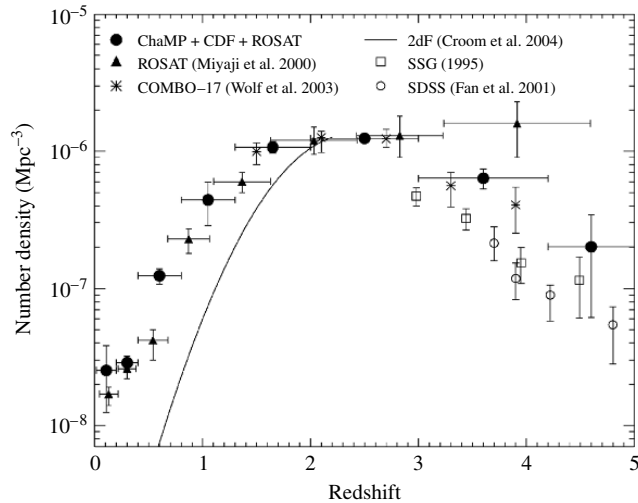


FIGURE 28.17 The comoving space density of active galactic nuclei (number per cubic megaparsec) as a function of redshift. (Figure adapted from Silverman et al., *Ap. J.*, 624, 630, 2005.)

dispersion of its spheroid increases, so does the mass of the central supermassive black hole. Evidence for this evolutionary relationship also appears in Fig. 26.17, which shows a Chandra X-ray image of two merging supermassive black holes in NGC 6240.

There is evidence, including interactions in observed quasars, suggesting that an individual quasar “event” lasts only for a galactic dynamical timescale (the dynamical timescale, essentially the characteristic free-fall or orbital time, was introduced in Section 26.2). Figure 28.18 shows the quasar PKS 2349–014 and a companion galaxy about the size of the LMC (the small bright spot just above the quasar). The thin curved wisps that almost surround the quasar are probably the result of a tidal interaction between the quasar and the companion galaxy. The companion is so close that it will probably merge with PKS 2349–014 in the near future. A portion of the diffuse nebulosity centered on the quasar may (or may not) represent a host galaxy in which it resides.

In 2003, James S. Dunlop, Ross J. McLure, and their colleagues reported on an extensive study of the morphology of the host galaxies of 33 radio-loud quasars, radio-quiet quasars, and radio galaxies in the redshift band $0.1 < z < 0.25$. The study was conducted by using the Hubble Space Telescope in combination with VLA radio imaging. The research team was able to conclude that all of the galaxies in their sample associated with radio-loud quasars or radio galaxies are massive ellipticals. Of the 13 radio-quiet quasars in the sample, 9 are hosted by massive ellipticals while the remaining 4 are in disk/bulge systems. Furthermore, of the 4 disk/bulge systems, the luminosities of 2 of them are dominated by their bulge components, implying that 11 of the 13 radio-quiet quasars (or $\sim 85\%$) are associated with galaxies that are predominantly spheroidal. In addition, the 2 disk-dominated galaxies are the sites of the lowest-luminosity AGNs in the sample by far, and they may be more appropriately considered Seyfert 2 galaxies. From the systems investigated in this study, it appears that all of the true quasars and radio galaxies in the sample are hosted by massive ellipticals that are nearly indistinguishable from lower- z quiescent galaxies typically found near the centers of rich clusters.

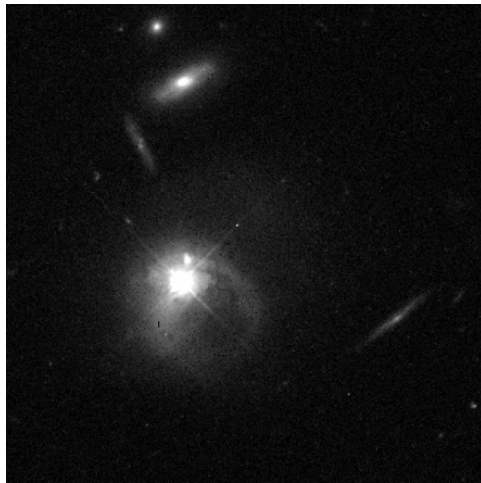


FIGURE 28.18 The quasar PKS 2349–014 in a gravitational interaction with a companion galaxy. (Figure from Bahcall, Kirhakos, and Schneider, *Ap. J. Lett.*, 447, L1, 1995. Courtesy of J. Bahcall, Institute for Advanced Study, NASA.)

The study also revealed that all of the radio-loud quasars contained central supermassive black holes of at least $10^9 M_{\odot}$ and that the radio-quiet quasars contained black holes with masses in excess of $5 \times 10^8 M_{\odot}$. It appears from this work that radio-loud systems are much less abundant than radio-quiet systems (10% vs. 90%) simply because the radio-loud systems require more massive central black holes to power the strong radio energy emission. However, although the investigation did find a broad correlation between increasing black hole mass and increasing radio luminosity, the most luminous radio sources cannot be attributed to black hole mass alone. Rather, there is speculation that black hole rotation also may be required to power the strongest radio sources.

In a second, statistical study of 12,698 quasars with redshifts in the range $0.1 < z < 2.1$ found in the Sloan Digital Sky Survey quasar catalog, McLure and Dunlop considered the evolution of black hole masses with increasing redshift. They found that black hole masses sufficient to power quasars were in place by $z \sim 2$. They further determined that all of the central black holes had masses in the range $10^7 M_{\odot} < M_{\text{bh}} < 3 \times 10^9 M_{\odot}$, where the upper limit corresponds to the most massive black holes yet found in the local universe (specifically, in M87 and Cygnus A).

The SDSS quasar study was also able to point out that quasar bolometric luminosities increase steadily with redshift from roughly $0.15L_{\text{Ed}}$ at $z \sim 0.2$ to $0.5L_{\text{Ed}}$ at $z \sim 2.0$, where L_{Ed} is the Eddington luminosity (Eq. 10.114). It is also evident from the data that the Eddington luminosity limit remains valid at the high- z end of the study.

Timescales of AGN Variability

The energy produced by many of the AGNs discussed above (excluding NLRGs and Seyfert 2s) can vary on short timescales. The luminosity of the broad emission lines and continuum of some Seyfert 1 galaxies and quasars can change by a factor of 2 within a few months, weeks, or even days, although there is little or no corresponding variation in the narrow lines. The variation in broad emission lines typically lags behind the continuum variation over similar timescales. There are also variations of a few percent in the visible and X-ray output of Seyfert 1s and quasars on timescales as short as a few minutes, with X-ray fluctuations typically the most rapid. At the other end of the scale, there may be changes of a longer duration. For example, Fig. 28.19 shows that around the year 1937, the quasar 3C 279 brightened by a factor of 250 at visible wavelengths during an outburst that lasted for several years.

Polarization of the Emission

Quasars typically show low degrees of polarization. At visible wavelengths, the degree of linear polarization is usually less than 3% for both radio-quiet and radio-loud objects, although it may reach up to 35% polarization for a few objects. The radio emission from high-polarization quasars comes mostly from a compact core; such quasars are called **core-dominant radio sources** (and also **compact sources**). These quasars are less polarized at radio wavelengths than the **lobe-dominant sources**, which may reach up to 60% linear polarization. The degree of linear polarization of AGN radio jets is typically 40% but may exceed 50% within a small region. (As discussed in Section 16.7, synchrotron radiation is highly linearly polarized. The lower polarization of the core-dominant sources is probably

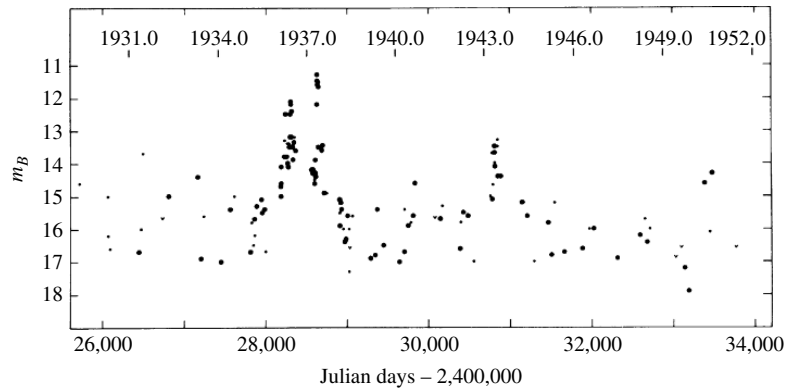


FIGURE 28.19 The variation in the apparent magnitude of the quasar 3C 279, based on an examination of archival astronomical photographs. (Figure adapted from Eachus and Liller, *Ap. J. Lett.*, 200, L61, 1975.)

due to synchrotron self-absorption.) Figure 28.20 shows the orientation of the magnetic field of the quasar 3C 47, as obtained from polarization measurements.

Fanaroff–Riley Luminosity Classes

In 1974, B. L. Fanaroff and J. M. Riley suggested that the radio-loud AGNs in the third Cambridge catalog could be categorized into two general luminosity classes. Fanaroff and Riley defined Class I objects as those for which the ratio of the distance between the brightest spots of radio emission on either side of the center (excluding the central source) to the full extent of the radio source is less than 0.5; Class II objects have a ratio greater than 0.5. An example of an FR I galaxy is NGC 1265, shown in Fig. 28.9, and Cyg A is a classic example of an FR II galaxy (Fig. 28.6). Quasars are also FR II objects.

From the classification scheme, it becomes apparent that FR I sources have diminishing radio luminosity with increasing distance from the center of the jets, while FR IIs tend to be most radio-bright at the ends of the lobes. It is also common that FR I galaxies have two recognizable radio jets, while FR II galaxies often exhibit only a single identifiable jet (the counterjet is either very weak or undetectable). Furthermore, FR I galaxies may have curved jets, while FR II jets tend to be straight.

Also intriguing, and the reason why this morphological classification scheme is referred to as a luminosity classification, is that there is also a rather clear demarcation between FR I and FR II classes in terms of the specific luminosity. Sources having a specific luminosity at 1.4 GHz of less than $10^{25} \text{ W Hz}^{-1}$ are identified as FR Is, and those with greater specific luminosities are inevitably classified as FR IIs.

Blazars

The properties of rapid variability and a high degree of linear polarization at visible wavelengths define the class of AGNs known as **blazars**. The most well-known object in this class is BL Lacertae, found in the northern constellation of Lacerta (the Lizard). BL Lac

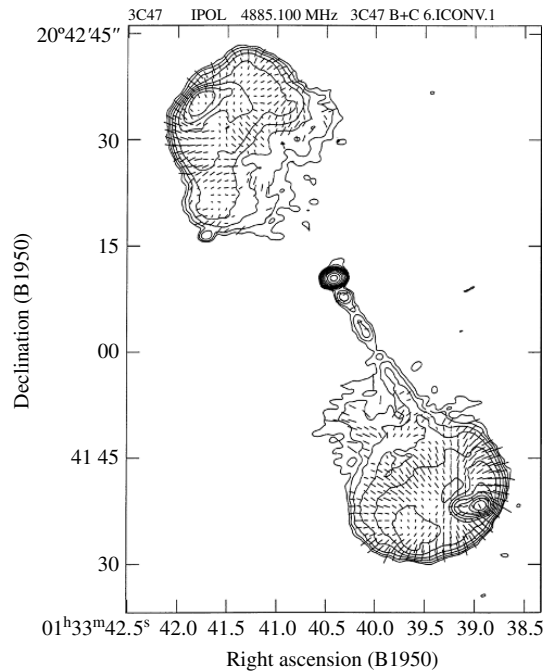


FIGURE 28.20 Polarization mapping of the magnetic field of the quasar 3C 47. Both lobes are highly polarized. (Figure adapted from Fernini et al., *Ap. J.*, 381, 63, 1991.)

was originally classified as a variable star because of its irregular variations in brightness; hence the variable star type of designation. In a week's time BL Lac would double its luminosity, and it would change by a factor of 15 as the months passed. But although BL Lac has a stellar appearance, its spectrum shows only a featureless continuum with very weak emission and absorption lines. Careful observations reveal that the bright, starlike nucleus of BL Lac is surrounded by a fuzzy halo that has a spectrum similar to that of an elliptical galaxy.

BL Lac objects are a subclass of blazars that are characterized by their rapid time-variability. Remarkably, their luminosities may change by up to 30% in just 24 hours and by a factor of 100 over a longer time period. BL Lacs are also distinguished by their strongly polarized power-law continua (30–40% linear polarization) that are nearly devoid of emission lines. However, observations of a few faint spectral lines have revealed high redshifts, so that, like quasars, BL Lacs are at cosmological distances. Of those BL Lacs that have been resolved, about 90% appear to reside in elliptical galaxies. Joining the BL Lac objects in the blazar classification are the **optically violently variable quasars** (OVVs). They are similar to the BL Lacs except that they are typically much more luminous, and their spectra may display broad emission lines.

LINERs

A final class of objects worth mentioning consists of the so-called **Low Ionization Nuclear Emission-line Regions** (LINERs). These galaxies have very low luminosities in their nuclei,

but with fairly strong emission lines of low-ionization species, such as the forbidden lines of [O I] and [N II]. The spectra of LINERs seem similar to the low-luminosity end of the Seyfert 2 class, and LINER signatures are detected in many (perhaps a majority of) spiral galaxies in high-sensitivity studies. These low-ionization lines are also detectable in starburst galaxies and in H II regions, and so it is unclear whether LINERs truly represent a low-luminosity limit of the AGN phenomena.

A Summary of AGN Classifications

This section has introduced a large number of objects with some commonalities and some apparent differences. Before proceeding to a discussion of how we might unify the description of AGN phenomena, we briefly summarize the objects in Table 28.1.

TABLE 28.1 A Summary of AGN Classes.

Class	Sub-class	Description
Seyferts	Type 1	broad and narrow emission lines, weak radio emission, X-ray emission, spiral galaxies, variable
	Type 2	narrow emission lines only, weak radio emission, weak X-ray emission, spiral galaxies, not variable
Quasars	Radio-loud (QSR)	broad and narrow emission lines, strong radio emission, some polarization, FR II, variable
	Radio-quiet (QSO)	broad and narrow emission lines, weak radio emission, weak polarization, variable
Radio Galaxies	BLRG	broad and narrow emission lines, strong radio emission, FR II, weak polarization, elliptical galaxies, variable
	NLRG	narrow emission lines only, strong radio emission, FR I and FR II, no polarization, elliptical galaxies, not variable
Blazars	BL Lacs	almost devoid of emission lines, strong radio emission, strong polarization, rapid variability, 90% in ellipticals
	OVV quasars	broad and narrow emission lines, strong radio emission, strong polarization, rapid variability, much more luminous than BL Lacs
ULIRGs		possibly dust-enshrouded quasars, alternatively may be starburst phenomena
LINERs		similar to low-luminosity Seyfert 2, low-ionization emission lines, in many (perhaps majority of) spiral galaxies, alternatively may be starburst phenomena or H II region emission

28.2 ■ A UNIFIED MODEL OF ACTIVE GALACTIC NUCLEI

The last section presented a bewildering array of observations of AGNs. Although there were many similarities, such as a bright compact nucleus, a wide continuum, and time variability, there were also many differences, including the presence or absence of broad emission lines, and the strength of radio and X-ray emission. The question is: Are the types of AGNs fundamentally different or fundamentally the same?

It now seems likely that active galactic nuclei are all powered by the same general engine, accretion onto central supermassive black holes (see page 959ff). Accordingly, the observational differences are due to the different orientations of the objects as viewed from Earth and to the different rates of accretion and masses of the central black holes. The presence of radio lobes is then something in addition to, and consistent with, the basic model.

Although there is not yet general agreement on some of the specific details of a unified model of AGNs, the model does serve to provide a framework for organizing the observations of AGNs and their interpretations. Any model should be self-consistent, meaning that its constitutive elements are all in harmony. Also, as with all viable theories and models in science, a successful AGN model should have the power to predict the results of new observational tests, in addition to explaining all of the observations made previously. Successful predictions have been made on the basis of the idea of unification, and it appears that the basic features of a unified model of AGNs are in hand. The purpose of this section is to demonstrate how the features of such a model may be deduced and to describe the best idea of what an AGN probably looks like.

Toward a Unified Model of AGNs

First, we will examine two pieces of evidence suggesting that the pursuit of a unified model is indeed justified. Figure 28.21 shows $L_{H\alpha}$, the luminosity in the $H\alpha$ emission line, and L_{FC} , the luminosity of the featureless continuum at a wavelength near 480 nm, plotted for a variety of AGNs (excluding blazars). If the hydrogen emission lines are produced via the photoionization of hydrogen atoms by the continuum radiation and the atoms' subsequent recombination, then the two luminosities should be proportional, and a straight line with a slope of 1 should be found on a log–log graph. The slope of the dashed line is 1.05, confirming that $L_{H\alpha} \propto L_{FC}$. This result implies a common origin for the hydrogen emission lines, both broad and narrow, that are observed in AGNs for Seyfert 1 and 2 galaxies, broad- and narrow-line radio galaxies, and radio-loud and radio-quiet quasars.

Another piece of evidence for a unified model comes from an observation reported by Robert Antonucci and Joseph Miller in 1985. When they observed NGC 1068 (a Seyfert 2) in polarized light, they found a Seyfert 1 spectrum with broad emission lines. This and similar cases discovered since then imply that within these Seyfert 2s are Seyfert 1 nuclei that are hidden from the direct view of Earth by some optically thick material. The diminished Seyfert 1 spectrum (normally overwhelmed by the direct Seyfert 2 spectrum) comes from light that reaches us indirectly by reflection from the interstellar medium outside the nucleus. This reflection would also contribute to the observed linear polarization, when the electric field vector is perpendicular to the radio axis. The orientation of the AGN relative to the line of sight from Earth will be an important factor in the unified model to be described.

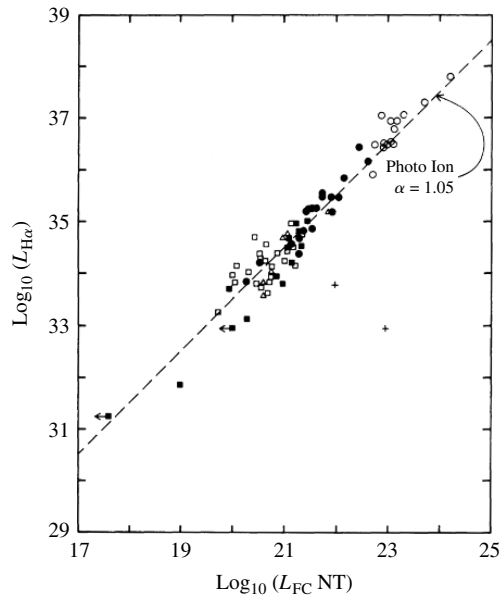


FIGURE 28.21 The luminosity in the $H\alpha$ emission line versus the luminosity of the featureless continuum at a wavelength near 480 nm (the “NT” stands for “nonthermal”). The symbols are quasars (open circles), Seyfert 1s (filled circles), Seyfert 2s (open squares), NLRGs (triangles), and more Seyfert 2s and NLRGs (filled squares). (Figure adapted from Shuder, *Ap. J.*, 244, 12, 1981.)

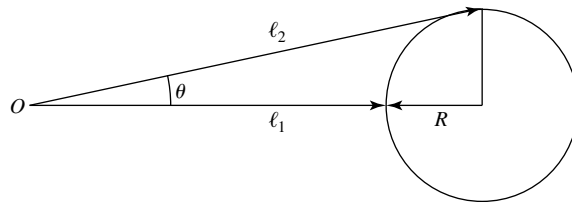


FIGURE 28.22 The brightening of a sphere as seen by a distant observer at point O .

The Nature of the Central Engine

The most important clue to the nature of the central engine that powers AGNs is their rapid time variability. Consider an optically thick sphere of radius R that simultaneously (in its own rest frame) brightens everywhere; see Fig. 28.22. The news of the change reaches a distant observer first from the nearest part of the sphere after traveling a distance ℓ_1 , and last from the edge or limb after traveling a distance ℓ_2 . (The back of the sphere isn’t seen.) Using

$$\ell_2 = \frac{\ell_1 + R}{\cos \theta} \simeq \ell_1 + R$$

for $R \ll \ell_1$ and $\cos \theta \simeq 1$, the light from the limb of the sphere must travel an additional distance of $\ell_2 - \ell_1 \simeq R$. The brightening is thus smeared out over a time interval $\Delta t = R/c$. In this way, the rapidity of a luminosity change can be used to set an upper limit on the size of the object involved. The high recessional speeds of quasars mean that relativity also plays a role. It is left as an exercise to show that if the sphere described above were moving away from Earth with a velocity v , then its radius as determined on Earth would be

$$R = c\Delta t \sqrt{1 - \frac{v^2}{c^2}} = \frac{c\Delta t}{\gamma}, \quad (28.4)$$

where γ is the Lorentz factor defined by Eq. (4.20). Using $\Delta t = 1$ hr for a typical value, and taking $\gamma = 1$ for convenience, the radius of the emitting region is no more than

$$R \simeq \frac{c\Delta t}{\gamma} = 1.1 \times 10^{12} \text{ m} = 7.2 \text{ AU}.$$

Considering that AGNs are the most luminous objects known, this is an incredibly small size. Whatever powers an active galactic nucleus would fit comfortably within our Solar System!

The typical quasar luminosity of $5 \times 10^{39} \text{ W}$ is equivalent to more than 360 Milky Way galaxies. However, there is an upper limit to the luminosity, L , of any spherically symmetric object that is in equilibrium. It must be less than the Eddington limit, $L < L_{\text{Ed}}$, where, from Eq. (10.114),

$$L_{\text{Ed}} \simeq 1.5 \times 10^{31} \text{ W} \left(\frac{M}{M_{\odot}} \right).$$

For a luminosity of $L = 5 \times 10^{39} \text{ W}$, this provides a lower limit for the mass:

$$M > \frac{L}{1.5 \times 10^{31} \text{ W}} M_{\odot} = 3.3 \times 10^8 M_{\odot}. \quad (28.5)$$

Finding such a large amount of mass in such a small space is clear evidence for a supermassive black hole. Using Eq. (17.27) for the Schwarzschild radius, the mass of a black hole with the radius R found in Eq. (28.4) is

$$M = \frac{Rc^2}{2G} = 3.7 \times 10^8 M_{\odot}.$$

The fact that these two mass estimates are of the same order of magnitude is enough to support the idea that supermassive black holes are involved in powering AGNs. For the rest of this section, we will assume a value of $10^8 M_{\odot}$ for a typical mass, which corresponds to a Schwarzschild radius of $R_S \simeq 3 \times 10^{11} \text{ m} \simeq 2 \text{ AU} \simeq 10^{-5} \text{ pc}$.

Generating Luminosity through Accretion

The most efficient way of generating energy is by the release of gravitational potential energy through mass accretion. As was shown in Example 18.1.2 for matter falling onto the surface of a $1.4 M_{\odot}$ neutron star, about 21% of the rest energy is released. However,

dropping matter straight down onto a black hole is very *inefficient* because there is no surface for the mass to strike. Instead, according to an observer at a great distance, a freely falling mass would slow to a halt and then disappear as it approached the Schwarzschild radius, R_S . On the other hand, as matter spirals in toward a black hole through an accretion disk, a substantial fraction of the rest energy can be released as viscosity converts kinetic energy into heat and radiation.

For a nonrotating black hole, the smallest stable circular orbit for a massive particle (and therefore the inner edge of an accretion disk) is at $r = 3R_S$. At this location, theoretical calculations show that the gravitational binding energy is 5.72% of the particle's rest-mass energy, so mass spiraling down through an accretion disk would release this much energy. The situation is even more favorable for a rotating black hole because the event horizon is located at a smaller r . For the most rapidly rotating black hole possible, both the event horizon and the smallest stable prograde orbit are at $r = 0.5R_S$ (the smallest stable retrograde orbit is at $4.5R_S$). The gravitational binding energy in this case of maximal rotation is calculated to be 42.3% of a particle's rest mass.

The **accretion luminosity** generated by a mass accretion rate, $\dot{M}$, through the disk may be written as

$$L_{\text{disk}} = \eta \dot{M} c^2, \quad (28.6)$$

where η is the efficiency of the process, $0.0572 \leq \eta \leq 0.423$. (For comparison, we found in Example 18.1.2 that the efficiencies for accretion onto a $0.85 M_\odot$ white dwarf and a $1.4 M_\odot$ neutron star are $\eta = 1.9 \times 10^{-4}$ and 0.21, respectively.)

The accretion of matter through a disk around a rapidly rotating black hole is an extremely efficient way of producing large amounts of energy. Furthermore, the smallest stable prograde orbit lies inside the ergosphere of a rapidly rotating black hole, and frame dragging guarantees that the accreting matter will rotate along with the black hole (recall Section 17.3). For these reasons, most astronomers believe that an accretion disk around a supermassive black hole is an essential ingredient of a unified model of AGNs. Figure 28.10 shows the spiral-shaped disk of gas that lies at the center of M87. The inner edge of the disk is rotating with a speed of about 550 km s^{-1} , causing the light from the lower right-hand edge of the disk to be blueshifted (approaching), while the light from the upper-right is redshifted (receding). The central supermassive black hole is calculated to have a mass of about $3 \times 10^9 M_\odot$.

Recall from Example 18.2.1 that the inner regions of accretion disks around white dwarfs and neutron stars are bright at ultraviolet and X-ray wavelengths, respectively. It might be expected that an accretion disk around a supermassive black hole would be a source of photons of even higher energies, but this is not the case. Because they are supported by degeneracy pressure, white dwarfs and neutron stars obey the mass–volume relation, Eq. (16.14), which states that these stars become smaller with increasing mass. Therefore, the accretion disks around more massive white dwarfs and neutron stars penetrate deeper into their gravitational potential wells. The Schwarzschild radius, however, increases with increasing mass, and so the characteristic disk temperature, T_{disk} (Eq. 18.20), decreases as the mass of the black hole increases. (Recall that the equation for T_{disk} was derived using Newtonian physics, but in this case a full relativistic treatment is clearly warranted.)

To see this, we will assume a rapidly rotating black hole and adopt $R = 0.5R_S = GM/c^2$ for the location of the inner edge of the accretion disk. Substituting R into Eq. (18.20), the characteristic disk temperature becomes

$$T_{\text{disk}} = \left(\frac{3c^6 \dot{M}}{8\pi\sigma G^2 M^2} \right)^{1/4}. \quad (28.7)$$

For a disk radiating at a fraction f_{Ed} of the Eddington limit,

$$f_{\text{Ed}} \equiv L_{\text{disk}}/L_{\text{Ed}}. \quad (28.8)$$

Equations (10.114) and (28.6) give

$$\eta \dot{M} c^2 = f_{\text{Ed}} \frac{4\pi G c}{\bar{\kappa}} M,$$

or

$$\dot{M} = \frac{f_{\text{Ed}}}{\eta} \frac{4\pi G}{\bar{\kappa} c} M. \quad (28.9)$$

Substituting this expression into Eq. (28.7) shows that

$$T_{\text{disk}} = \left(\frac{3c^5 f_{\text{Ed}}}{2\bar{\kappa}\sigma G M \eta} \right)^{1/4}, \quad (28.10)$$

and so for the disk temperature, $T_{\text{disk}} \propto M^{-1/4}$.

Example 28.2.1. Consider an accretion disk around a rapidly rotating supermassive black hole of $10^8 M_\odot$. The value of f_{Ed} is probably close to 1 for luminous quasars and roughly between 0.01 and 0.1 for Seyfert galaxies. In this example, let the disk luminosity be equal to the Eddington limit ($f_{\text{Ed}} = 1$), so from Eq. (10.114), $L = 1.5 \times 10^{39}$ W. Also, we will adopt $\eta = 0.1$ as a representative accretion efficiency. The mass accretion rate required to maintain the disk luminosity is

$$\dot{M} = \frac{f_{\text{Ed}}}{\eta} \frac{4\pi G}{\bar{\kappa} c} M = 1.64 \times 10^{23} \text{ kg s}^{-1} = 2.60 M_\odot \text{ yr}^{-1}.$$

Luminous quasars must be fed at a rate of around 1 to $10 M_\odot \text{ yr}^{-1}$. Less luminous AGNs may have correspondingly smaller appetites. The characteristic disk temperature is

$$T_{\text{disk}} = \left(\frac{3c^5 f_{\text{Ed}}}{2\bar{\kappa}\sigma G M \eta} \right)^{1/4} = 7.30 \times 10^5 \text{ K},$$

where Eq. (9.27) with $X = 0.7$ has been used for the opacity due to electron scattering. According to Wien's displacement law (Eq. 3.19), the spectrum of a blackbody with this temperature peaks at a wavelength of 39.7 nm, in the extreme ultraviolet region of the electromagnetic spectrum.

Although this expression for T_{disk} is at best a rough estimate of the characteristic disk temperature, temperatures of several hundred thousand kelvins agree with the results of more realistic disk calculations.

It is thought that the big blue bump observed in the spectra of quasars is the thermal signature of an underlying accretion disk. However, although accretion is believed to provide the power for AGNs, the theoretical spectrum of an accretion disk (such as that shown in Fig. 18.14) cannot account for the wide continuum that is actually observed.

Structure of the Accretion Disk

A detailed model of the accretion disk around a supermassive black hole is difficult to derive because the high luminosities involved must have a significant effect on the disk's structure. Theoretical calculations indicate that the structure of the accretion disk depends on f_{Ed} (Eq. 28.8). Several possible structures have been identified. If $f_{\text{Ed}} < 0.01$, then the density of the disk is too small for efficient cooling. The energy generated by the disk's viscosity (internal friction) cannot be radiated away efficiently, and the disk puffs up into an **ion torus** that is supported by the pressure of the hot ions. Part or all of the disk would then resemble a doughnut around the central black hole. Values of $0.01 < f_{\text{Ed}} < 0.1$ or so imply a geometrically thin disk, as described in Section 18.2 for close binary systems (recall that, by definition, at any radial distance r in a thin disk, the vertical height $h \ll r$). As the value of L_{disk} becomes super-Eddington ($f_{\text{Ed}} > 1$), the radiation pressure that is generated balances the force of gravity and the photons are capable of supporting the matter in an inflated **radiation torus**.

One scenario involves a composite disk that has three regions, as shown in Fig. 28.23. Within about $1000R_S$ of the center, radiation pressure exceeds the gas pressure, resulting in a thick, hot disk. This is the probable origin of the big blue bump in the continuous spectrum. Exterior to this, reaching out to some $10^5 R_S$ ($\simeq 1$ pc for $M = 10^8 M_\odot$), is a thin disk that is supported by gas pressure. This part of the disk flares outward, becoming thicker with increasing radius. The concave surface of the outer disk means that it can be irradiated by the central source or the thick, hot portion of the inner disk, resulting in a wind flowing

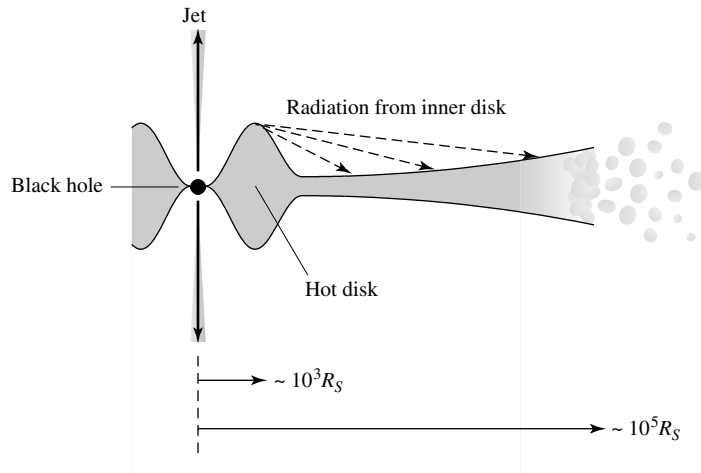


FIGURE 28.23 A schematic structure of the accretion disk in an AGN. The radial direction is not drawn to scale.

outward from the disk. Finally, beyond about $10^5 R_S$, the thick disk breaks up into numerous small clouds.

Unfortunately, some problems with this picture remain. For example, the values of f_{Ed} quoted above for Seyferts appear to be incompatible with models of thick disks.

The Implications of AGN Spectra

For the blazars, at least, there is widespread agreement that the continuous spectrum, with its power-law form and significant polarization, is produced by synchrotron radiation. As we saw in Section 16.7 for pulsars, synchrotron radiation is produced when relativistic charged particles, such as electrons, spiral around magnetic field lines. (For objects other than blazars, the situation is more complicated. Recall that the big blue bump observed in the continua of other types of AGNs is believed to be thermal radiation. In addition, dust emission plays an important role in the infrared.)

With a wide range of photon energies available for ionizing atoms, synchrotron radiation can account for the variety of ionization states observed in the emission line spectra of AGNs. For example, a number of ionization states have been seen for forbidden lines, including [O I] and [Fe X]. Furthermore, synchrotron radiation can be up to 70% linearly polarized, in agreement with the high degrees of polarization observed for some AGNs.

Producing a Relativistic Outflow of Charged Particles

It might seem surprising to find a magnetic field involved in this situation. Although it is theoretically possible for an isolated black hole to have a magnetic field, it is unlikely to occur naturally. This is because although the three attributes of a black hole (its mass, angular momentum, and electric charge) can be combined to produce a magnetic field, black holes should be essentially electrically neutral, since any net charge acquired by a black hole would be rapidly canceled as it attracted charge of the opposite sign. However, the ionized disk material is highly conducting, so there can be a magnetic field that is generated by the accretion disk as the disk orbits the black hole.

It may be that, in a manner reminiscent of the mechanism described for pulsars in Section 16.7, the varying magnetic field near the surface of the disk induces a large electric field that is capable of accelerating charged particles away from the disk. As the particles move outward, they are accelerated to relativistic speeds while they spiral around the magnetic field lines that rotate with the disk. Because the field lines are anchored to the conducting disk, the particle energy ultimately comes at the expense of the accretion energy.

There is another source of energy that taps the rotational energy of the black hole itself, first described by Roger Blandford and Roman Znajek. Detailed calculations show that the rotating black hole can be thought of as a spinning conductor in a magnetic field; see Fig. 28.24. Just as the motion of a conducting wire through a magnetic field will produce an electromotive force (emf) between its ends, the rotation of a black hole in a magnetic field will produce a potential difference between its poles and its equator. The effective resistance of the rotating black hole between its poles and equator is about 30Ω . In this picture, the black hole acts like an immense battery connected to a $30\text{-}\Omega$ resistor. Power is extracted from the black hole's rotational energy, just as if a current of charged particles were flowing through the resistor in response to the voltage difference. As a result, the black

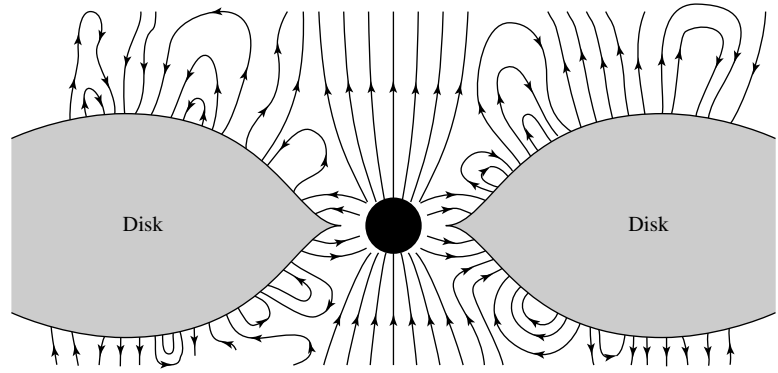


FIGURE 28.24 An accretion disk and its magnetic field orbiting a rotating black hole.

hole's rate of spin is reduced. The power generated by the **Blandford–Znajek mechanism** is approximately

$$\begin{aligned}
 P &\simeq \frac{4\pi}{\mu_0} B^2 R_S^2 c \\
 &= 2.7 \times 10^{38} \text{ W} = 7.1 \times 10^{11} L_\odot
 \end{aligned}
 \tag{28.11}$$

for a $10^8 M_\odot$ black hole with $R_S = 3 \times 10^{11} \text{ m}$ and a magnetic field of 1 T. The energy is in the form of electromagnetic radiation and a flow of relativistic pairs of electrons and positrons. Up to 9.2% of the rest energy of a maximally rotating black hole may be extracted in this manner. This is another important source of energy, comparable to that obtained by disk accretion.

The two processes just described appear capable of producing a relativistic outflow of charged particles, although the mechanism is still uncertain. As the electrons spiral around the magnetic field lines, they emit the synchrotron radiation that contributes to the continuous spectrum of AGNs. (You will recall from Section 28.1 that the observed power-law synchrotron spectra of AGNs imply that the distribution of electron energies is described by a power law. Just how this distribution comes about is as yet unknown, but power-law synchrotron spectra are also observed for supernova remnants such as the Crab Nebula.)

The Generation of X-Rays

AGNs can be very bright in X-rays,⁸ and several mechanisms are usually invoked to explain the excess over that produced directly by synchrotron radiation. The high-frequency end of the accretion disk spectrum may be sufficient to account for the soft (low-energy) X-rays. Lower-energy photons from other sources may also be scattered to much higher

⁸X-rays from AGNs are probably responsible for the *X-ray background*, a uniform glow of X-ray photons that pervades the sky. Discovered in 1962, the X-ray background extends into the gamma-ray region of the electromagnetic spectrum.

energies by collisions with relativistic electrons. As the name suggests, this **inverse Compton scattering** is the reverse of the Compton scattering process described in Section 5.2. In addition, inverse Compton scattering may produce the gamma rays coming from the quasar 3C 273. Thermal bremsstrahlung, the mechanism that produces the X-ray emission observed in clusters of galaxies, has a characteristic spectrum (Fig. 27.17) that could also be consistent with observations of X-rays from AGNs.

Broad-Line and Narrow-Line Emission

The characteristic broad emission lines (when present) and narrow emission lines of AGNs are the result of photoionization by the continuum radiation. A careful examination of these lines reveals much about the conditions under which they were formed. All of the broad lines arise from permitted atomic transitions, but none of them involve the forbidden transitions seen in some narrow lines. The broad $H\alpha$ and $H\beta$ lines vary on timescales of a month or less, while the narrow lines seem to vary little, if at all. This evidence, along with the discovery that Seyfert 2 galaxies may harbor Seyfert 1 nuclei that are hidden from direct view by some obscuring material, indicates that the broad and narrow lines in the spectra of AGNs originate in different regions under different conditions.

The Broad-Line Region

The broad emission lines observed in the spectrum of many AGNs are formed in a **broad-line region** that is relatively close to the center. A study of the Seyfert galaxy NGC 4151 revealed that when the intensity of the continuum radiation varied, most of the broad emission lines responded very quickly, within a month or less and perhaps as quickly as one week. Light can travel a distance of nearly 10^{15} m in 30 days, so this provides a rough estimate of the radius of the broad-line regions for Seyferts and BLRGs. The variation of the lines in quasars takes place more slowly, so their broad-line regions may be larger by a factor of four or so. A study of the broad Fe II emission lines that are usually present indicates that the temperature in the broad-line region is $\sim 10^4$ K. Other lines indicate that the number density of electrons probably lies between 10^{15} m^{-3} and 10^{16} m^{-3} . Forbidden lines will not be seen with large number densities such as these because of the frequent collisions between the atoms. Atoms and ions with electrons in the long-lived metastable states that give rise to forbidden lines are deexcited by collisions before downward radiative transitions can occur. As a result, the forbidden lines are much weaker than the permitted lines.

There is widespread agreement that the broad-line region must be clumpy, containing partially ionized clouds of gas, rather than being homogenous. The optically thick clouds that actually produce the emission lines fill only about 1% of the available volume and probably have a flattened distribution. These regions of high density may be surrounded by a rarefied, high-temperature medium that prevents the clouds from dispersing.

According to the unified model, the various types of observed AGN phenomena (see Table 28.1) derive from different viewing angles of the central engine and surrounding environment. The unified model postulates that a large, optically thick torus of gas and dust surrounds the clouds of the broad-line region. This is presumably what conceals the broad-line region and the central source from direct view when observing a Seyfert 2 galaxy; see Fig. 28.25. In this case, the continuum and emission lines must reach the observer indirectly

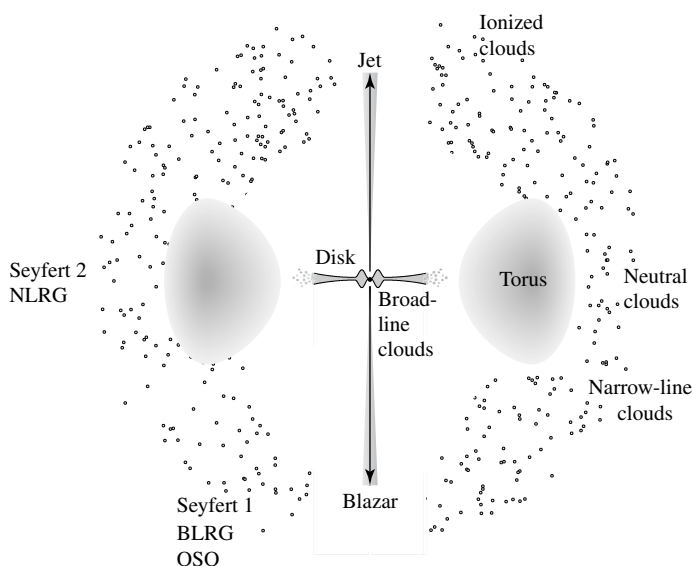


FIGURE 28.25 A sketch of a unified model of an active galactic nucleus. The jets would be present in a radio-loud AGN. A typical observer's point of view is indicated for AGNs of various types.

by reflected light, which explains why the continuum of Seyfert 2s is much fainter than the continuum of Seyfert 1s. Overall, the light received directly from the central nucleus makes Seyfert 1 galaxies generally brighter than Seyfert 2s. The torus is also thought to be opaque to soft X-rays because the X-rays observed for Seyfert 2s are hard (with energies above approximately 5 keV).

Determining Black Hole Masses in Broad-Line Regions

The broad emission lines also indicate that the clouds orbit the central supermassive black hole. In fact, taking the 5000 km s^{-1} width of the lines to be an orbital velocity and using $r = 10^{15} \text{ m}$ as an orbital radius provides an estimate of the central mass. From Eq. (24.49),

$$M_{\text{bh}} = \frac{rv^2}{G} = 1.9 \times 10^8 M_{\odot},$$

which is consistent with previous mass estimates.

A second technique for determining the masses of the central black holes of AGNs in broad-line regions is based on measuring the lag time between changes in brightness of the continuum and emission lines. This **reverberation mapping** technique combines the measured time delay, τ , with the root-mean-square width of the emission line, σ_{line} , giving a mass estimate of

$$M_{\text{bh}} = \frac{fc\tau\sigma_{\text{line}}^2}{G}, \quad (28.12)$$

where c is the speed of light and f is a factor that depends on the structure, kinematics, and

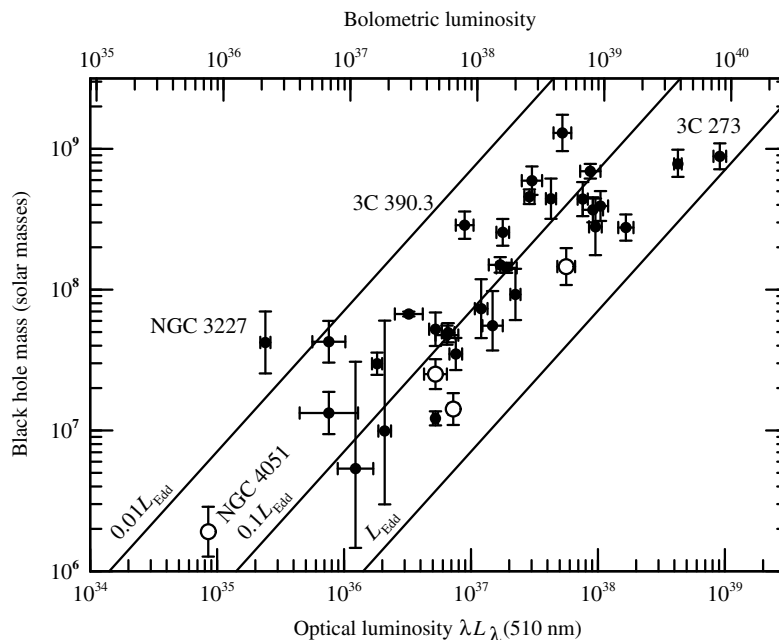


FIGURE 28.26 The masses of the supermassive black holes in AGNs as a function of their luminosities. The solid diagonal lines represent lines of constant fractions of the Eddington luminosity. The masses were determined by the reverberation mapping technique. The bolometric luminosities are given on the top axis, and the optical luminosities (centered at 510 nm) are given on the bottom axis. (Figure courtesy of Bradley Peterson.)

orientation of the broad-line region. Clearly $c\tau$ is a measure of the size of the broad-line region. Of course it is important to be consistent in terms of which ionized species are used in making the measurement (in other words, which emission line is used in the study) because highly ionized species tend to have the shortest lag times, suggesting that the level of ionization in the broad-line region is dependent on distance from the central source.

By comparison with other black hole–spheroid relations, such as the mass–velocity dispersion relation displayed in Fig. 25.15 for a sample of resolved galactic centers, the scaling factor in Eq. (28.12) is found to have the value $f \approx 5.5$. The reverberation mapping method requires long-term observations to determine the lag times, and high spectral resolution, but it does not require spatially resolving the central region of the host galaxy. Thus, the reverberation technique holds great promise for measuring black hole masses in AGNs that are at high z . Figure 28.26 shows the results of a study of AGN broad-line emission regions.

The Narrow-Line Region

Outside the opaque torus is the **narrow-line region** where the narrow emission lines originate. The number density of electrons in the narrow-line region is only about 10^{10} m^{-3} , comparable to the values found in planetary nebulae and dense H II regions. The narrow-line region contains more mass than the broad-line region, and both permitted and forbidden lines can be formed in such an environment. They reveal a temperature of approximately

10^4 K. Like the broad-line region, the region that generates the narrow-line spectrum is clumpy. It is probably composed of a more or less spherical distribution of clouds. The clouds that are far enough above or below the plane of the obscuring torus can be illuminated and photoionized by the continuum radiation from the center. Other clouds have their lines of sight to the central source blocked by the opaque torus, and so they remain neutral.

In fact, if the narrow-line region can be treated as a clumpy H II region, then Eq. (12.32) for the Strömgen radius can be used to estimate the fraction of the narrow-line region that is occupied by clouds. If the clouds occupy a fraction ϵ (referred to as the **filling factor**) of the volume of the narrow-line region, then Eq. (12.32) can be modified to produce an estimate of the radius of that region,

$$r_{\text{NLR}} \approx \left(\frac{3N}{4\pi\alpha_{\text{qm}}\epsilon} \right)^{1/3} \frac{1}{n_e^{2/3}}. \quad (28.13)$$

In this case N is the number of photons per second produced by the central source of the AGN that have enough energy to ionize hydrogen from the ground state, and α_{qm} is a quantum-mechanical recombination coefficient (*not* the spectral index), as given on page 432.

Example 28.2.2. To estimate the filling factor of the narrow-line region, we will assume an AGN luminosity of $L = 5 \times 10^{39}$ W. The continuum includes photons with a wide range of energies. We will assume that the monochromatic energy flux (Eq. 28.1) obeys a power law with a spectral index of $\alpha = 1$. Recalling Example 28.1.1, the flux is related to the luminosity by

$$L = 4\pi d^2 \int_{\nu_1}^{\nu_2} F_\nu d\nu = \int_{\nu_1}^{\nu_2} C \nu^{-1} d\nu$$

with $\nu_1 = 10^{10}$ Hz and $\nu_2 = 10^{25}$ Hz for the range of frequencies of the continuous spectrum (cf., Fig. 28.14), and C is a constant to be determined.⁹ Evaluating the integral and solving for C gives

$$C = \frac{L}{\ln(\nu_2/\nu_1)} = \frac{L}{\ln 10^{15}} = 0.029L.$$

We are now ready to find N , the number of photons emitted per second with an energy $E_H > 13.6$ eV, or a frequency $\nu_H > 3.29 \times 10^{15}$ Hz, required to ionize hydrogen from the ground state. Dividing the monochromatic energy flux by the energy per photon, $E_{\text{photon}} = h\nu$, results in

$$N = \int_{\nu_H}^{\nu_2} \frac{C \nu^{-1}}{h\nu} d\nu = \int_{\nu_H}^{\nu_2} \frac{0.029L}{h\nu^2} d\nu \simeq \frac{0.029L}{h\nu_H} = 6.64 \times 10^{55} \text{ s}^{-1},$$

where $\nu_2 \gg \nu_H$.

continued

⁹The constant C includes the leading factor of $4\pi d^2$.

Observations of the nearest Seyfert 2 galaxies show narrow-line regions with diameters between roughly 100 and 1000 pc. If we let $r_{\text{NLR}} = 200$ pc, $n_e = 10^{10} \text{ m}^{-3}$, and $\alpha_{\text{qm}} = 3.1 \times 10^{-19} \text{ m}^3 \text{ s}^{-1}$ (as on page 432), the filling factor of the narrow-line region is approximately

$$\epsilon \approx \frac{3N}{4\pi\alpha_{\text{qm}}} \frac{1}{n_e^2 r_{\text{NLR}}^3} = 2.2 \times 10^{-2}.$$

Thus clouds occupy roughly 2% of the volume of the narrow-line region.

The profiles of the narrow emission lines seen in Seyfert 2s often have extended blue wings, indicating that the clouds are moving toward us relative to the galactic nucleus. This is usually interpreted as a radial flow of the clouds away from the center. The light from the clouds moving away from us on the far side of the AGN is presumably diminished by extinction.

An outward flow of clouds in the region that produces narrow emission lines could be driven by a combination of radiation pressure and a wind coming from the accretion disk, as mentioned previously, or the outflow could be associated with the material in radio jets. Figure 28.27 shows an image obtained by the Hubble Space Telescope of the narrow-line region

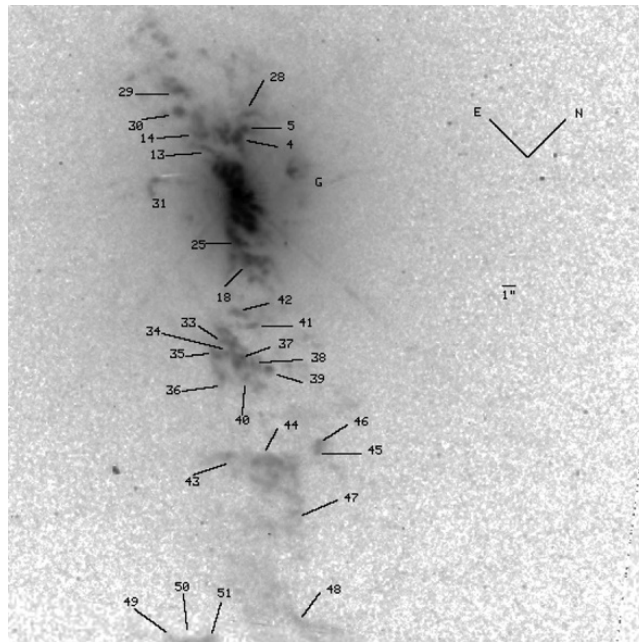


FIGURE 28.27 An HST image of the narrow-line region of the Seyfert 1 galaxy, NGC 4151. Numerous clouds are evident in a biconical distribution. The clouds to the southwest are approaching the observer relative to the nucleus, and the clouds to the northeast have recessional velocities. There is some evidence that the clouds may be associated with the galaxy's radio jets. An angular scale of $1''$ is indicated on the image; at the distance to NGC 4151, $1''$ corresponds to a projected linear distance of 63 pc. The labels correspond to clouds identified in the paper. (Figure from Kaiser et al., *Ap. J.*, 528, 260, 2000.)

region for the Seyfert 1 galaxy, NGC 4151. Distinct emission clouds are clearly evident in this high-resolution image. It is also evident that the optical emission falls within two conical distributions extending to the northeast and the southwest of the center of the galaxy. When a radio map is overlaid on the image, the radio emission also falls along the same axis as the biconical optical emission.

Similar observations have been made of Mrk 3, a Seyfert 2 galaxy. In this case there is additional evidence that the narrow-line region is composed of expanding shells around the radio jets. It has been suggested that the material forming the radio jets is expanding at near $0.1c$ from the center of the galaxy. As the jets move through the interstellar material, the gas is ionized at a temperature of $\sim 10^7$ K. The overheated gas expands outward away from the jets, energizing the gas near the surface of the expanding shell, which then produces the narrow-line emission region.

A further consequence of the expanding jet model of producing the narrow-line region is that the region may be relatively short-lived, at least in the case of the relatively weak Seyfert systems. Since the radio jets of Seyfert galaxies are typically only a few kiloparsecs in length, this implies that ages of roughly 10^4 to 10^5 years can be deduced based on the expansion velocity of the jets. If that is the case, Seyfert phenomena may be relatively transient events, perhaps caused by the temporary feeding of the galaxy's supermassive black hole.

A Summary of the Unified Model of AGNs

The preceding details comprise what must be considered as a rough sketch of a unified model of an AGN. Its central engine is an accretion disk orbiting a rotating, supermassive black hole. The AGN is powered by the conversion of gravitational potential energy into synchrotron radiation, although the rotational kinetic energy of the black hole may also serve as an important energy source. The structure of the accretion disk depends on the ratio of the accretion luminosity to the Eddington limit. To supply the observed luminosities, the most energetic AGNs must accrete between about 1 and $10 M_{\odot} \text{ yr}^{-1}$. The perspective of the observer, together with the mass accretion rate and mass of the black hole, largely determines whether the AGN is called a Seyfert 1, a Seyfert 2, a BLRG, a NLRG, or a radio-loud or radio-quiet quasar.

Although many of the details of the unified model have not yet been fully confirmed, the unified model does appear to provide an important framework for describing many of the general characteristics of active galaxies. For example, Fig. 28.28 shows an amazing HST image of NGC 4261, an elliptical radio galaxy in the Virgo cluster that is classified optically as a LINER. The core of this radio-loud object shows a bright nucleus surrounded by a large, obscuring torus that is perpendicular to the radio jets. The central object is probably a $10^7 M_{\odot}$ black hole, although the HST image does not have the resolution to confirm this. The torus has a radius of about 70 pc (2×10^{18} m), and the jets reach out some 15 kpc from the nucleus.¹⁰

¹⁰The Schwarzschild radius of a $10^7 M_{\odot}$ black hole is 2.95×10^{10} m, smaller than the torus by nearly eight orders of magnitude. The black hole would be less than a nanometer wide on the scale of Fig. 28.28.

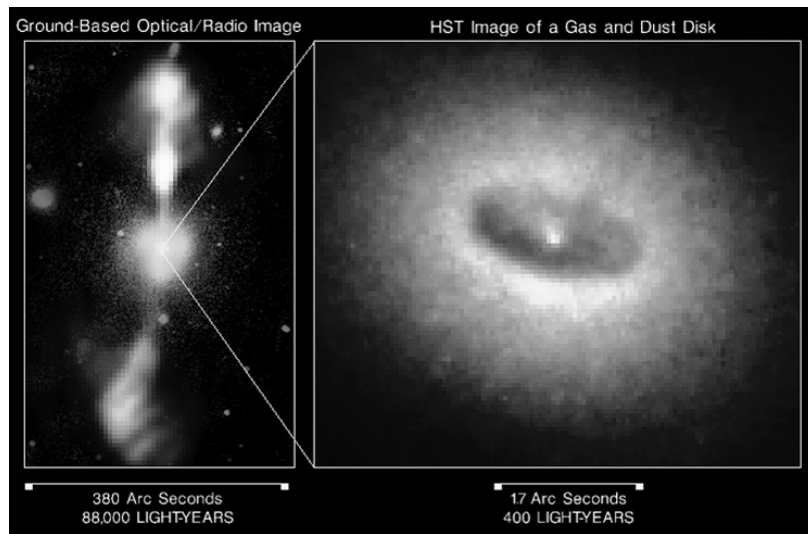


FIGURE 28.28 Two views of NGC 4261. Left: a composite optical and radio image from the ground, showing the radio jets. Right: an optical image from the HST, showing the dusty torus around the nucleus. (Courtesy of NASA.)

28.3 ■ RADIO LOBES AND JETS

There is a basic division of active galaxies into objects that are radio-loud and those that are radio-quiet. Radio-loud sources usually consist of a radio core, one or two detectable jets, and two dominant radio lobes. The radio-quiet sources are less luminous at radio wavelengths by a factor of 10^3 to 10^4 , consisting of a weak radio core and perhaps a feeble jet. The increased level of activity in radio-loud AGNs is not confined to radio wavelengths, however; they also tend to be about three times brighter in X-rays than their radio-quiet cousins.

The Generation of Jets

The radio lobes are produced by jets of charged particles ejected from the central nucleus of the AGN at relativistic speeds; recall Fig. 28.8. These particles are accelerated away from the nucleus in two opposite directions, powered by the energy of accretion and/or by the extraction of rotational kinetic energy from the black hole via the Blandford–Znajek mechanism. The jet must be electrically neutral overall, but it is not clear whether the ejected material consists of electrons and ions or an electron–positron plasma. The latter, being less massive, would be more easily accelerated. The disk’s magnetic field is coupled (“frozen in”) to this flow of charged particles. The resulting magnetic torques may remove angular momentum from the disk, which would allow the accreting material to move inward through the disk.

The incredible narrowness and straightness of some jets means that a collimating process must be at work very near the central engine powering the jet. A thick, hot accretion

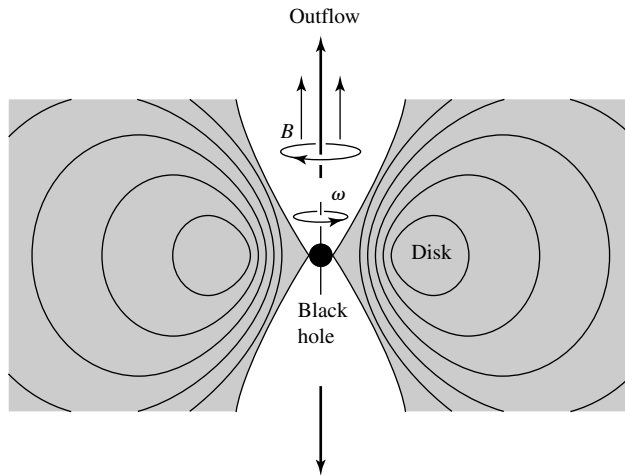


FIGURE 28.29 A schematic showing the collimation of outflowing material by a thick, hot accretion disk. The loops represent contours of constant disk density.

disk around the black hole could provide natural collimation by funneling the outflowing particles, as shown in Fig. 28.29. Because the accreting material retains some angular momentum as it spirals inward through the disk, it will tend to pile up at the smallest orbit that is compatible with its angular momentum. Inside this “centrifugal barrier” there may be a relatively empty cavity that can act as a nozzle, directing the accreting gases outward along the walls of the cavity. However, producing highly relativistic jets, as frequently observed, appear to be difficult to accomplish with this nozzle mechanism.

Alternatively, magnetohydrodynamic (MHD) effects could play an important role in accelerating and collimating the relativistic flows. Unfortunately, details of MHD mechanisms have not yet been fully developed either. Whatever the specific details of the collimation of jets, their straightness is likely to be linked to the rotating supermassive black holes at the hearts of AGNs.

The Formation of Radio Lobes

As a jet of material travels outward, its energy primarily resides in the kinetic energy of the particles. However, the jet encounters resistance as it penetrates the interstellar medium within the host galaxy and the intergalactic medium beyond. As a result, the material at the head of the jet is slowed, and a shock front forms there. The accumulation and deceleration of particles at the shock front cause the directed energy of the jet to become disordered as the particles “splash back” to form a large lobe in which the energy may be shared equally by the kinetic and magnetic energy. The problem of calculating the motion of a jet through the intergalactic medium is so complicated that extensive numerical simulations are required to model the process. Figure 28.30 shows a series of computer simulations of jets with various initial energies working their way through the intergalactic medium.

The motion of the charged particles and the magnetic fields within the lobes of radio-loud objects contain an enormous amount of energy. For Cygnus A, the energy of each

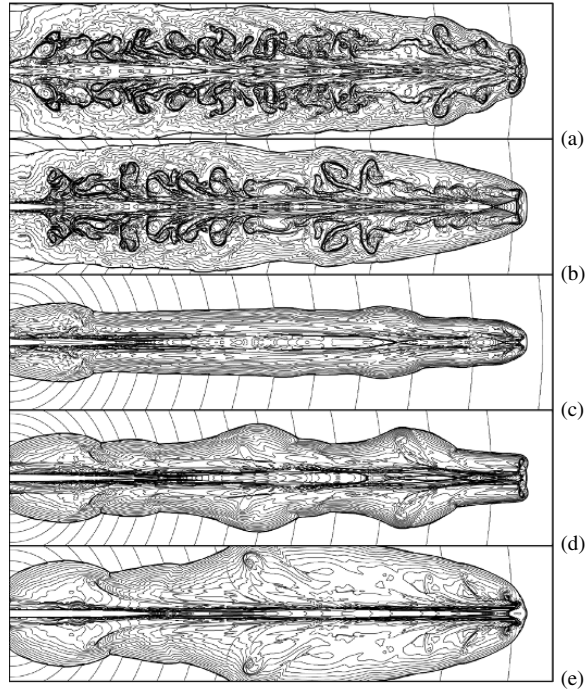


FIGURE 28.30 Numerical simulation of electron–positron plasma jets moving through the intergalactic medium, which is assumed to be decreasing in density with increasing distance from the source of the jets (left-hand side of each frame). The frames correspond to initial Lorentz factors (γ) at the source of the jets of (a) 2.0, (b) 2.5, (c) 5.0, (d) 7.0, and (e) 10.0. Somewhat different behaviors are seen in the simulations when the jet material is assumed to be composed of electrons and protons. (Figure from Carvalho and O’Dea, *Ap. J. Suppl.*, 141, 371, 2002.)

lobe is estimated to be approximately 10^{53} to 10^{54} J, equivalent to the energy liberated by 10^7 supernovae.

Example 28.3.1. Assuming that each radio lobe of Cyg A contains an energy of $E_{\text{lobe}} = 10^{53}$ J, and adopting $h = [h]_{\text{WMAP}} = 0.71$ for the values given in Example 28.1.1 for Cyg A, the lifetime of the radio lobes can be estimated. With Cyg A’s radio luminosity of $L_{\text{radio}} = 4.8 \times 10^{37}$ W, the time to radiate away the energy stored in its radio lobes is

$$t_{\text{lobe}} = \frac{E_{\text{lobe}}}{L_{\text{radio}}} = 66 \text{ Myr.}$$

Generally, the lifetime of the radio emission from radio lobes ranges from 10^7 to more than 10^8 years.

The average strength of the magnetic field in the lobes can be estimated by making the common assumption that the energy is shared equally between the kinetic and magnetic

energy. From Eq. (11.9), the magnetic energy stored per unit volume is $u_m = B^2/2\mu_0$. If the volume of the lobe is V_{lobe} , then

$$\frac{1}{2} E_{\text{lobe}} = u_m V_{\text{lobe}} = \frac{B^2 V_{\text{lobe}}}{2\mu_0}$$

or

$$B = \sqrt{\frac{\mu_0 E_{\text{lobe}}}{V_{\text{lobe}}}}. \quad (28.14)$$

Example 28.3.2. Assume that each of Cyg A's radio lobes can be modeled as a sphere of radius $R = 8.5 \text{ kpc} = 2.6 \times 10^{20} \text{ m}$, characteristic of the size of the lobes. With $E_{\text{lobe}} = 10^{53} \text{ J}$, the average value of the magnetic field in the lobes is estimated to be

$$B = \sqrt{\frac{\mu_0 E_{\text{lobe}}}{\frac{4}{3}\pi R_{\text{lobe}}^3}} \approx 41 \text{ nT}.$$

A value of order 10 nT is typical of the bright emission regions ("hot spots" that are a few kpc across) found in radio lobes. In diffuse radio lobes, the value may be more than an order of magnitude smaller, while the field strength in the radio core is probably around 100 nT.

Accelerating the Charged Particles in the Jets

The observations of jets are made possible by inefficiencies in the transport of particles and energy out to the radio lobes. The spectra of the radio lobes and jets follow a power law, with a typical spectral index of $\alpha \simeq 0.65$. The presence of power-law spectra and a high degree of linear polarization strongly suggest that the energy emitted by the lobes and jets comes from synchrotron radiation.

The loss of energy by synchrotron radiation is unavoidable, and in fact the relativistic electrons in jets will radiate away their energy after just 10,000 years or so. This implies that there is not nearly enough time for particles to travel out to the larger radio lobes; for example, for the large radio galaxy 3C 236, the journey would take several million years, even at the speed of light. This long travel time and the long lifetime of radio lobes imply that there must be some mechanism for accelerating particles in the jets and radio lobes. As one possibility, shock waves may accelerate charged particles by magnetically squeezing them, reflecting them back and forth inside the shock. Radiation pressure may also play a role, but it alone is not enough to generate the necessary acceleration.

Superluminal Velocities

Although the standard model of jets and radio lobes requires a steady supply of charged particles moving at relativistic speeds, evidence for such high velocities is difficult to obtain. The absence of spectral lines in a power-law spectrum means that the relativistic velocity of the jet material cannot be measured directly but must be inferred from indirect evidence. The most compelling argument for relativistic speeds involves radio observations of material

ejected from the cores of several AGNs with so-called **superluminal velocities**. This effect is observed within about 100 pc of the AGN's center and probably continues farther out.

Example 28.3.3. Figure 28.31 is a radio view of the core of the quasar 3C 273 that shows a blob of radio emission moving away from the nucleus with an angular velocity of $\mu = 0.0008'' \text{ yr}^{-1}$. Assuming that the radio knot is traveling in the plane of the sky, perpendicular to the line of sight, and using a distance of $d = 440h^{-1} \text{ Mpc}$ for 3C 273, the *apparent* transverse velocity of the blob away from the nucleus is, from Eq. (1.5),

$$v_{\text{app}} = d\mu = 1.67 \times 10^9 h^{-1} \text{ m s}^{-1} = 5.57h^{-1}c.$$

If $h = [h]_{\text{WMAP}}$, we find that $v_{\text{app}} = 7.85c$. This is clearly unphysical, and so the assumption of motion perpendicular to the line of sight must be wrong.

Figure 28.32 shows how the motion of the knot toward the observer can resolve this dilemma. Suppose a source is traveling with a velocity v (the actual speed of the source, not its apparent speed) at an angle ϕ measured from the line of sight. A photon is emitted along the line of sight at time $t = 0$ when the source is a distance d from Earth. At a later time (t_e), another photon is emitted when the distance to Earth is $d - vt_e \cos \phi$. The first

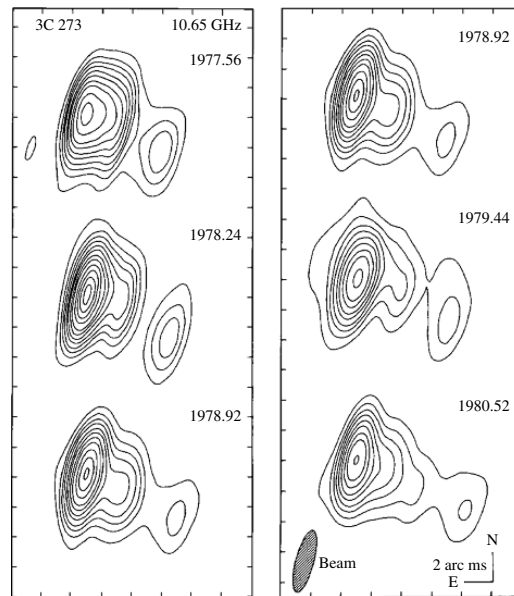


FIGURE 28.31 The motion of a radio-emitting knot ejected from the core of the quasar 3C 273. The dates of the observations are recorded as fractions of a year, and the third image has been repeated for clarity. (Figure adapted from Pearson et al., *Nature*, 290, 365, 1981. Reprinted by permission from *Nature*, Vol. 290, pp. 365–368. Copyright 1981 Macmillan Magazines Limited.)

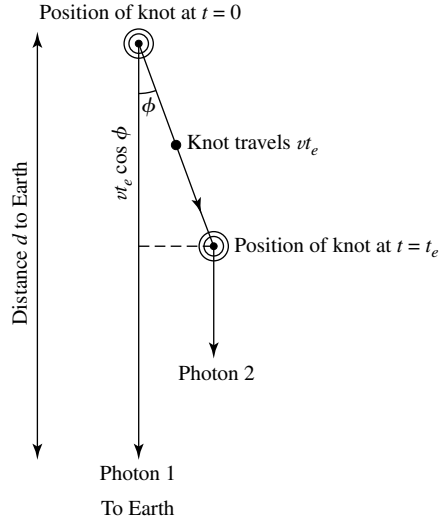


FIGURE 28.32 Two photons emitted at $t = 0$ and $t = t_e$ by a source moving with speed v .

photon reaches Earth at time t_1 , where

$$t_1 = \frac{d}{c}.$$

The second photon arrives at Earth at time

$$t_2 = t_e + \frac{d - vt_e \cos \phi}{c}.$$

The time on Earth between the reception of the two photons is thus

$$\Delta t = t_2 - t_1 = t_e \left(1 - \frac{v}{c} \cos \phi \right),$$

a time that is *shorter* than t_e . The apparent transverse velocity measured on Earth is then

$$v_{\text{app}} = \frac{vt_e \sin \phi}{\Delta t} = \frac{v \sin \phi}{1 - (v/c) \cos \phi}.$$

Solving this for v/c results in

$$\boxed{\frac{v}{c} = \frac{v_{\text{app}}/c}{\sin \phi + (v_{\text{app}}/c) \cos \phi}.} \quad (28.15)$$

It is left as an exercise to show that $v/c < 1$ for angles satisfying

$$\frac{v_{\text{app}}^2/c^2 - 1}{v_{\text{app}}^2/c^2 + 1} < \cos \phi < 1, \quad (28.16)$$

and that the smallest possible value of v/c for the source is

$$\frac{v_{\min}}{c} = \sqrt{\frac{v_{\text{app}}^2/c^2}{1 + v_{\text{app}}^2/c^2}}, \quad (28.17)$$

which occurs at an angle $\phi_{\min}$ given by

$$\cot \phi_{\min} = \frac{v_{\text{app}}}{c}. \quad (28.18)$$

This minimum value of v/c corresponds to a minimum Lorentz factor (Eq. 4.20) of the source of

$$\gamma_{\min} = \frac{1}{\sqrt{1 - v_{\min}^2/c^2}} = \sqrt{1 + v_{\text{app}}^2/c^2} = \frac{1}{\sin \phi_{\min}}. \quad (28.19)$$

Example 28.3.4. Referring to Example 28.3.3, since the actual speed of the radio knot ejected by 3C 273 must be less than c , it is necessary from Eq. (28.16) that

$$\phi < \cos^{-1} \left(\frac{v_{\text{app}}^2/c^2 - 1}{v_{\text{app}}^2/c^2 + 1} \right) = 14.5^\circ.$$

That is, the knot must be approaching Earth within 14.5° of the line of sight. From Eq. (28.17), the lower limit of the knot's speed is $v_{\min} = 0.992c$. Therefore, from Eq. (28.19), $\gamma_{\min} = 7.92$.

The minimum value of the Lorentz factor inferred for other superluminal sources ranges between $\gamma_{\min} = 4$ and 12 for $h = [h]_{\text{WMAP}}$. 3C 273 and similar examples provide compelling evidence that the central cores of AGNs can accelerate material to relativistic speeds.

Relativistic Beaming and One-Sided Jets

The headlight effect described in Section 4.3 will be involved whenever a source of light moves with a relativistic speed ($\gamma \gg 1$). All of the light emitted into the forward hemisphere in the rest frame of the source is concentrated into a narrow cone in the observer's rest frame. The cone's half-angle, θ , is given by Eq. (4.43), $\sin \theta = 1/\gamma$. Comparing this with Eq. (28.19) above shows that if the source is approaching Earth with a relativistic velocity within the angle $\phi_{\min}$ of the line of sight, this relativistic beaming effect will cause it to appear much brighter than expected and it will appear to be moving with a superluminal speed across the plane of the sky. Interestingly, nearly all AGNs showing superluminal motions are surrounded by large, dim halos that may be radio lobes seen end-on. Blazars may be quasars or radio galaxies viewed with the jet coming directly (or nearly so) toward the observer. Their very rapid time variability could then be exaggerated by the relativistic Doppler shift; see Eq. (4.31). Any luminosity variations due to a source within the relativistic jet would be observed to occur approximately 2γ times more rapidly by astronomers on Earth (see Problem 28.12).

Conversely, a relativistic source moving away from us will appear unusually dim (recall Fig. 28.6). All of the jets showing superluminal motion are one-sided, even when the

AGNs exhibit two radio lobes. It is expected that the central engines of AGNs produce two oppositely directed jets; however, relativistic beaming seems to explain why the jets appear to be only one-sided.

The Roles of Galactic Companions

The galactic companions of AGNs may play an important role in supplying them with the fuel. Most Seyferts (at least 90%) are spiral galaxies, and many have close neighbors with whom they may be interacting. Gravitational perturbations produce the distorted appearance frequently seen in those Seyferts close enough to be studied, as evidenced by the appearance of bars and/or outer rings. As described in Section 26.1, these interactions could produce gravitational torques on the gas in a Seyfert galaxy, drastically reducing its angular momentum and sending the gas plunging into the galactic center. The result would be the delivery of a fresh supply of fuel to the Seyfert nucleus to be accreted by the black hole. The concentration of gas could also result in a burst of star formation around the nucleus. Furthermore, if a merger with a galactic companion occurs, the subsequent disruption could produce an elliptical galaxy with an active nucleus, resulting in a young radio galaxy.

Mergers are certainly important for quasars as well. Some low-redshift quasars show evidence of past interactions (see Fig. 28.18), and mergers were undoubtedly more common in the early universe than they are today. Since galaxies are believed to have contained more gas when they were young, mergers may have resulted in the infall of large amounts of gas that could have contributed to the growth of a central supermassive black hole as the gas simultaneously fueled its activity. In addition, mergers probably resulted in the coalescing of supermassive black holes, producing even larger central engines. As the masses of the black holes grew, so did the number of quasars and their energy output, until the fuel powering the engines was largely consumed.

AGN Evolution

What happens when a quasar runs out of fuel? In broad terms, the diminishing fuel supply of an energetic object could lead to its transformation into a less luminous form. For example, Cen A has huge radio lobes (see Fig. 28.11) but is a weak radio source. It was probably much more luminous in the past but is now fading away. On the other hand, a lesser luminosity could be explained by a less massive black hole rather than a smaller accretion rate. Our Milky Way does not have a $10^8 M_{\odot}$ black hole at its center, although there is a more modest one of $3.7 \times 10^6 M_{\odot}$ (Section 24.4). If, as has been conjectured, every large galaxy comparable to the Milky Way has a supermassive black hole of at least $10^6 M_{\odot}$, then low-level galactic activity may be a common occurrence.

One large impediment to understanding the evolution of active galaxies is our current lack of knowledge about their lifetimes. Some researchers find that around $z = 2$, the number of luminous AGNs decreases toward the present-day epoch with a characteristic decay time of $\tau \simeq 2h^{-1}$ Gyr. However, this is only an *upper limit* to the lifetime of an AGN. A single AGN may remain active this long, or the individual lifetimes may be much shorter, say between 10^7 and 10^8 years, the typical timescale needed to radiate away the energy stored in a radio lobe. In this latter case, τ would describe the statistical changes in a population of active galaxies, rather than the behavior of a single individual. A galaxy may then experience just

one, or several, brief episodes of activity during its history as mergers refuel the central engine. It may be that Seyfert galaxies experience recurring episodes of activity, for example (recall the discussion on page 1121).

28.4 ■ USING QUASARS TO PROBE THE UNIVERSE

Quasars are among the most distant visible objects in the universe, and so they offer a unique opportunity to probe the intervening space. Clouds of gas, galaxies, and dark matter can all affect the light from a quasar during its journey to Earth. By decoding the clues supplied by observations of quasars, astronomers can learn a great deal about the perturbing objects along the line of sight.

Gravitational Lensing and Multiple Images of Quasars

In 1919, the same year that Eddington measured the bending of starlight as it passed near the Sun and verified Einstein's general theory of relativity (see Section 17.1), Sir Oliver Lodge (1851–1940), an English physicist, put forth the possibility of using a gravitational lens to focus starlight (recall the discussion of gravitational lensing; see Fig. 24.14). During the 1920s, astronomers began to consider how light passing through the curved spacetime surrounding a massive object could produce multiple images of the source. Then, in 1937, Fritz Zwicky (1898–1974) proposed that gravitational lensing by a galaxy would be much more likely than gravitational lensing by individual stars. By the 1970s the search was on for a multiply imaged quasar, and in 1979 the quasar Q0957+561 was discovered to appear twice in the sky. As shown in Fig. 28.33, the two images are separated by $6.15''$, and each

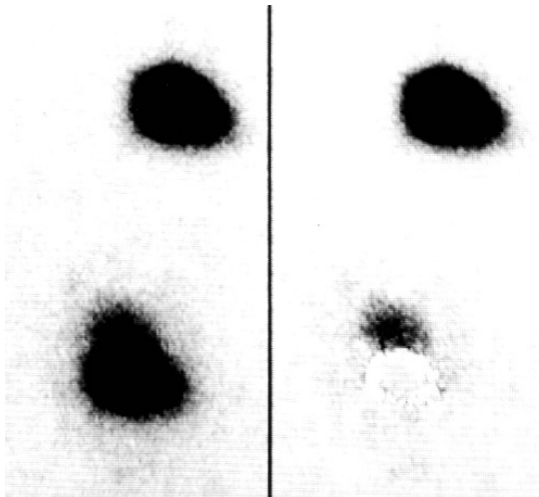


FIGURE 28.33 An optical (negative) view of the double quasar Q0957+561. The photo on the left shows the two gravitationally lensed images. The fuzz extending upward from the bottom image is the lensing galaxy. On the right, the upper image has been subtracted from the lower image to reveal the lensing galaxy more clearly. (Figure from Stockton, *Ap. J. Lett.*, 242, L141, 1980.)